数据库查询：过程与截图

NHLI70037 ICA resit 2026 · Bob · SCN1A 变异（题目写法 c.3993C>T，p.Asp1331Glu）
查询时间：2026-11-04 23:30–23:57 UTC（每张图的来源栏有精确访问时间）　|　文档生成：2026-11-05
浏览器：云端容器内 Chromium（Playwright 驱动），不是你的 Chrome（本会话没有 Claude in Chrome）。黄色高亮、红色圆角框和顶部来源栏是我加的，页面内容未改动。
NCBI 网页（nuccore、ClinVar、PubMed、PMC）对该浏览器弹出图片验证码，按规则未点击、未绕过；相关查询改用 NCBI E-utilities 官方接口在浏览器中显示，或用等效公开数据库（gnomAD、VariantValidator、Mutalyzer、UniProt）代替。

0. 结论汇总

#	问题	结果	依据
1	题目写法 c.3993C>T (p.Asp1331Glu) 是否成立	不成立。c. 与 p. 互相矛盾（第 3 位 C>T 只能把 GAC 变成 GAT，仍是 Asp）；MANE 转录本 c.3993 参考碱基为 A，第 1331 位为 Glu	D02–D07
2	参考转录本	NM_001165963.4 = MANE Select（= ENST00000674923.1），编码 NP_001159435.1（2009 aa，与 UniProt P35498 序列一致）	D01、D06
3	最可能的真实变异（推断）	NM_001165963.4:c.3993A>T 或 c.3993A>C → p.(Glu1331Asp)；GRCh38 chr2:166009728 T>A / T>G（GRCh37 166866238）	D02、D06
4	gnomAD v4.1.2	该位点没有任何变异（已勾选 Indels 和 Filtered variants）；覆盖良好；相邻位点 AN ≈ 1,605,000	D08–D11
5	ClinVar	第 1331 位无记录；邻近 1328、1329、1333 位有 P/LP 错义	D12、D13
6	旁系同源对应位点	SCN2A p.Glu1321Gln = LP（1 个提交者）；SCN8A p.Glu1311Val = VUS；SCN3A 无记录	D14、D29
7	基因层面	Dravet、GEFS+、DEE 均 Definitive；HI score 3；错义 Z = 8.78；pLoF o/e = 0.07（pLI = 1）	D01、D15、D16
8	蛋白位置	E1331 位于 DIII S4 跨膜螺旋（1314–1332）的胞内端，紧邻 S4–S5 linker	D17
9	计算预测（p.Glu1331Asp）	REVEL 0.613（Pejaver 2022 校准的不确定区间：PP3 和 BP4 都不适用）；AlphaMissense 0.85；CADD 23–26；SpliceAI 全部 0.00	D07、D18–D21
10	保守性（手动比对）	E 在 9 个人类 Nav 旁系同源和四足动物直系同源中保守；斑马鱼 scn1laa 为 D，果蝇 para 为 Q	表 A1
11	ClinGen VCEP SCN1A 细则 v2.0	PM1 适用（PER #10 含 1331）；PM2 仅 Supporting；PP2、PP4 不适用；PP3 用 REVEL 且封顶 Moderate；PS2/PM6 按表型计分	D22–D29

A. 命名与参考序列核对

推导（不依赖任何数据库）：密码子 n 占 c.(3n−2)～c.3n；3993 = 3 × 1331，所以 c.3993 是第 1331 号密码子的第 3 位。Asp 的密码子是 GAT/GAC，Glu 是 GAA/GAG（图 D03）。第 3 位 C>T 只能把 GAC 变成 GAT（仍是 Asp，同义）；Asp→Glu 必须是第 3 位变成 A 或 G。因此 “c.3993C>T (p.Asp1331Glu)” 在任何转录本上都不可能同时成立。

A1　参考转录本是 MANE Select

gnomAD 基因页列出 MANE Select = ENST00000674923.1 / NM_001165963.4（D01）；VariantValidator 结果也标注 Transcript Flag: MANE Select（D06）。【需查】NCBI RefSeq 页面（KEYWORDS “MANE Select”）因验证码未能截图；我用 E-utilities 下载的 GenBank 记录核对过：KEYWORDS = “RefSeq; MANE Select.”，CDS

479..6508，转录本第 4469–4471 位 = gaa（c.3991–3993）。需要的话请在你的 Chrome 打开 https://www.ncbi.nlm.nih.gov/nuccore/NM_001165963.4 补截。

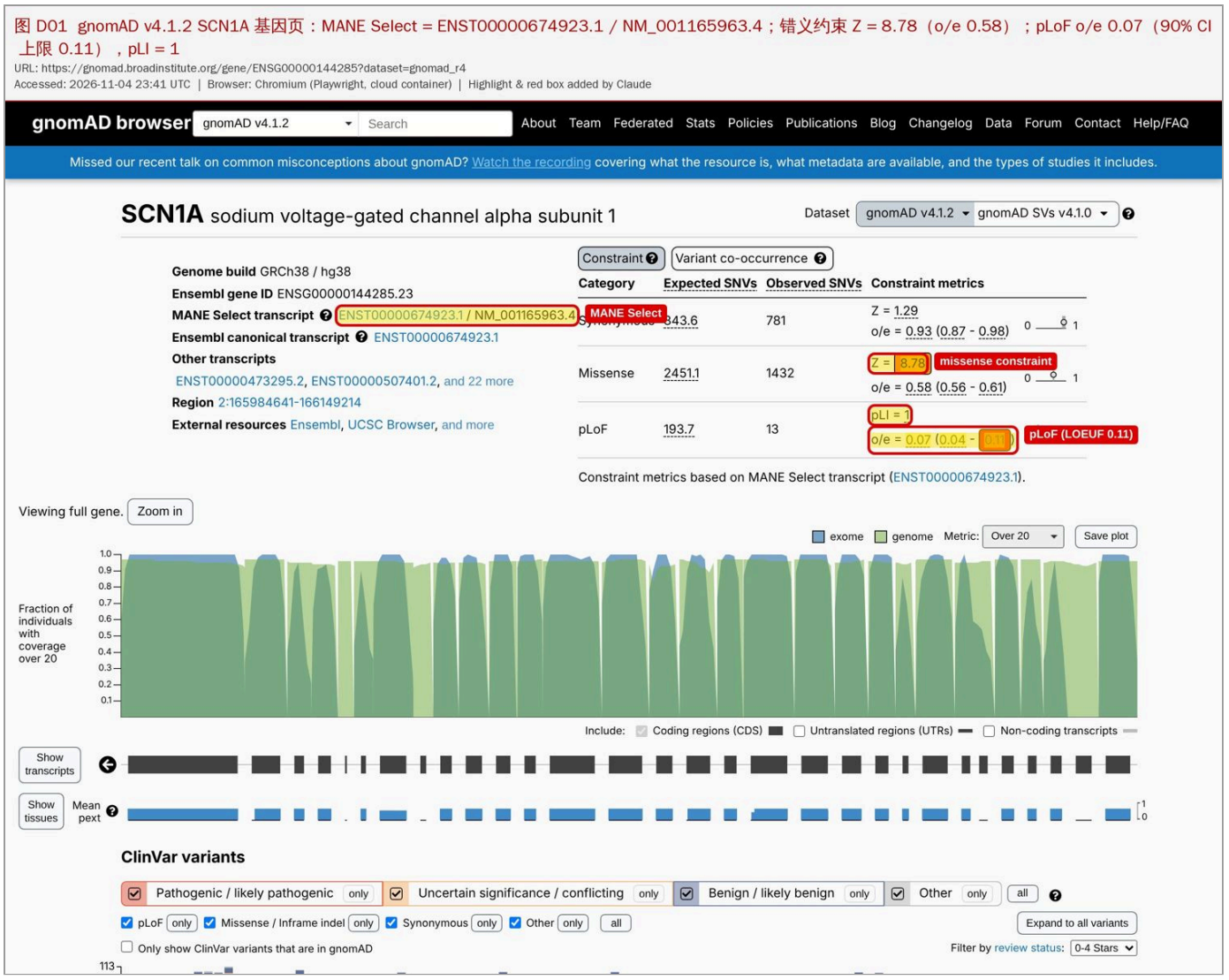


图 D01　gnomAD v4.1.2 SCN1A 基因页：MANE Select = ENST00000674923.1 / NM_001165963.4；错义约束 Z = 8.78（o/e 0.58）；pLoF o/e 0.07（90% CI 上限 0.11），pLI = 1

文件：screenshots/D01_gnomAD_MANE与基因约束.jpg ｜ 来源：https://gnomad.broadinstitute.org/gene/ENSG00000144285?dataset=gnomad_r4 ｜ 访问：2026-11-04 23:41 UTC

A2　遗传密码

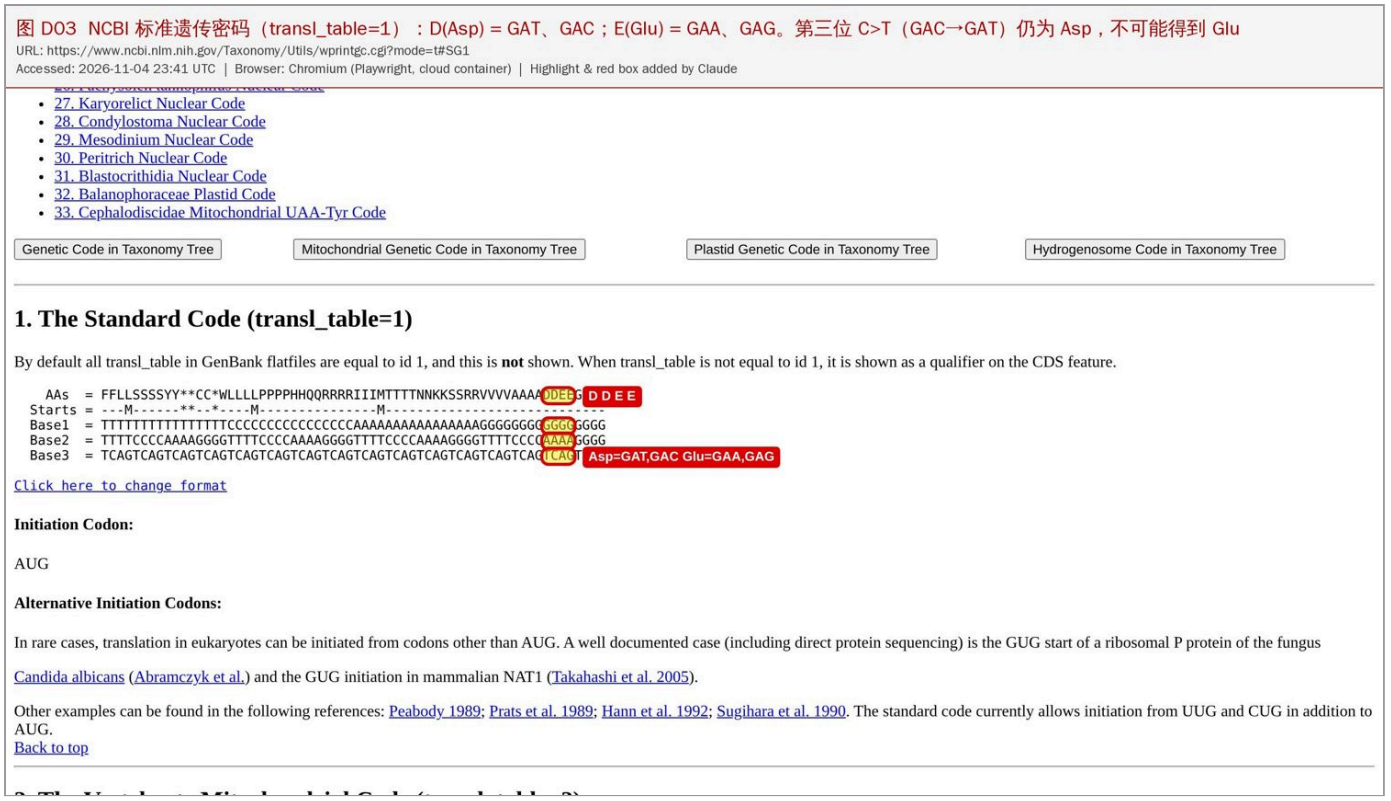


图 D03　NCBI 标准遗传密码（transl_table=1）：D(Asp) = GAT、GAC；E(Glu) = GAA、GAG。第三位 C>T（GAC→GAT）仍为 Asp，不可能得到 Glu

文件：screenshots/D03_NCBI_标准遗传密码表_Asp_Glu.jpg ｜ 来源：https://www.ncbi.nlm.nih.gov/Taxonomy/Utils/wprintgc.cgi?mode=t#SG1 ｜ 访问：2026-11-04 23:41 UTC

A3　题目写法校验：两个独立工具都报参考碱基不符

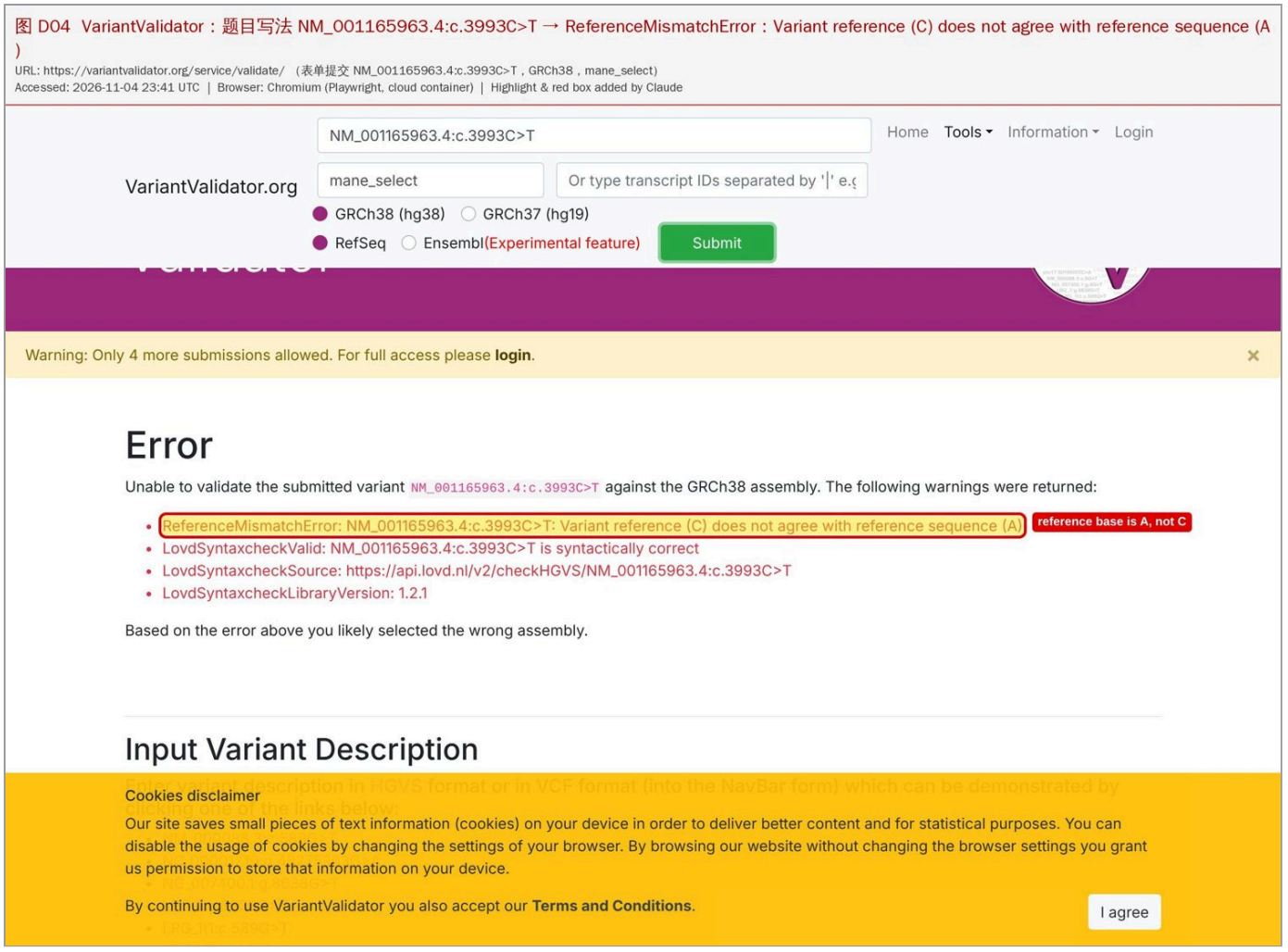


图 D04　VariantValidator：题目写法 NM_001165963.4:c.3993C>T → ReferenceMismatchError：Variant reference (C) does not agree with reference sequence (A)

文件：screenshots/D04_VariantValidator_c3993CT_参考碱基不符.jpg　|　来源：https://variantvalidator.org/service/validate/ （表单提交 NM_001165963.4:c.3993C>T，GRCh38，mane_select）　|　访问：2026-11-04 23:41 UTC



图 D05　Mutalyzer 3（独立第二个工具）：c.3993C>T → "`C` was not found in the reference sequence; `A` was found instead."

文件：screenshots/D05_Mutalyzer_c3993CT_参考碱基不符.jpg ｜ 来源：https://mutalyzer.nl/normalizer/NM_001165963.4:c.3993C%3ET ｜ 访问：2026-11-04 23:42 UTC

A4　推断写法校验：c.3993A>T 成立，给出 p.(Glu1331Asp) 和基因组坐标

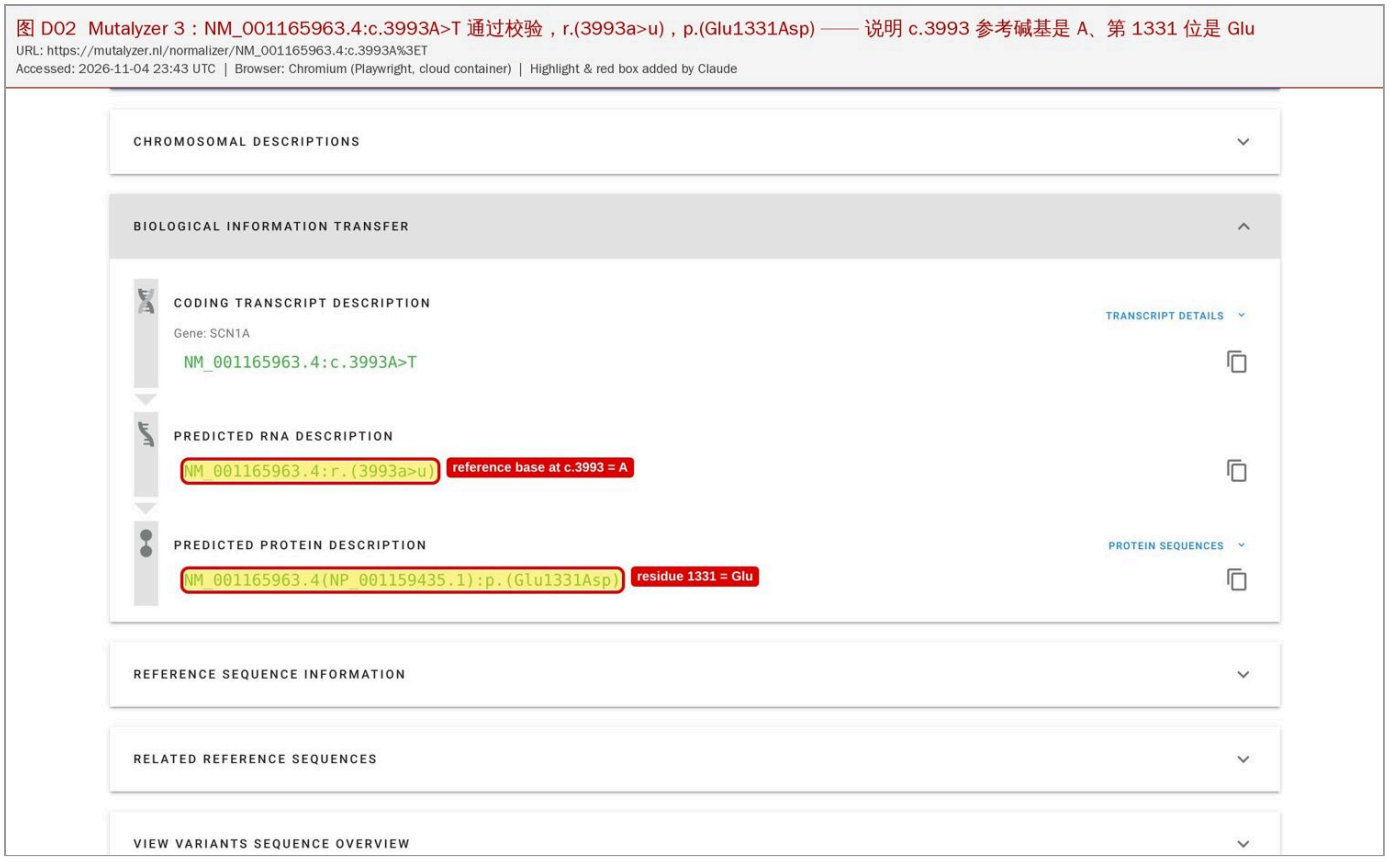


图 D02　Mutalyzer 3：NM_001165963.4:c.3993A>T 通过校验，r.(3993a>u)，p.(Glu1331Asp) —— 说明 c.3993 参考碱基是 A、第 1331 位是 Glu

文件：screenshots/D02_Mutalyzer_c3993AT_参考碱基A_Glu1331.jpg ｜ 来源：https://mutalyzer.nl/normalizer/NM_001165963.4:c.3993A%3ET ｜ 访问：2026-11-04 23:43 UTC

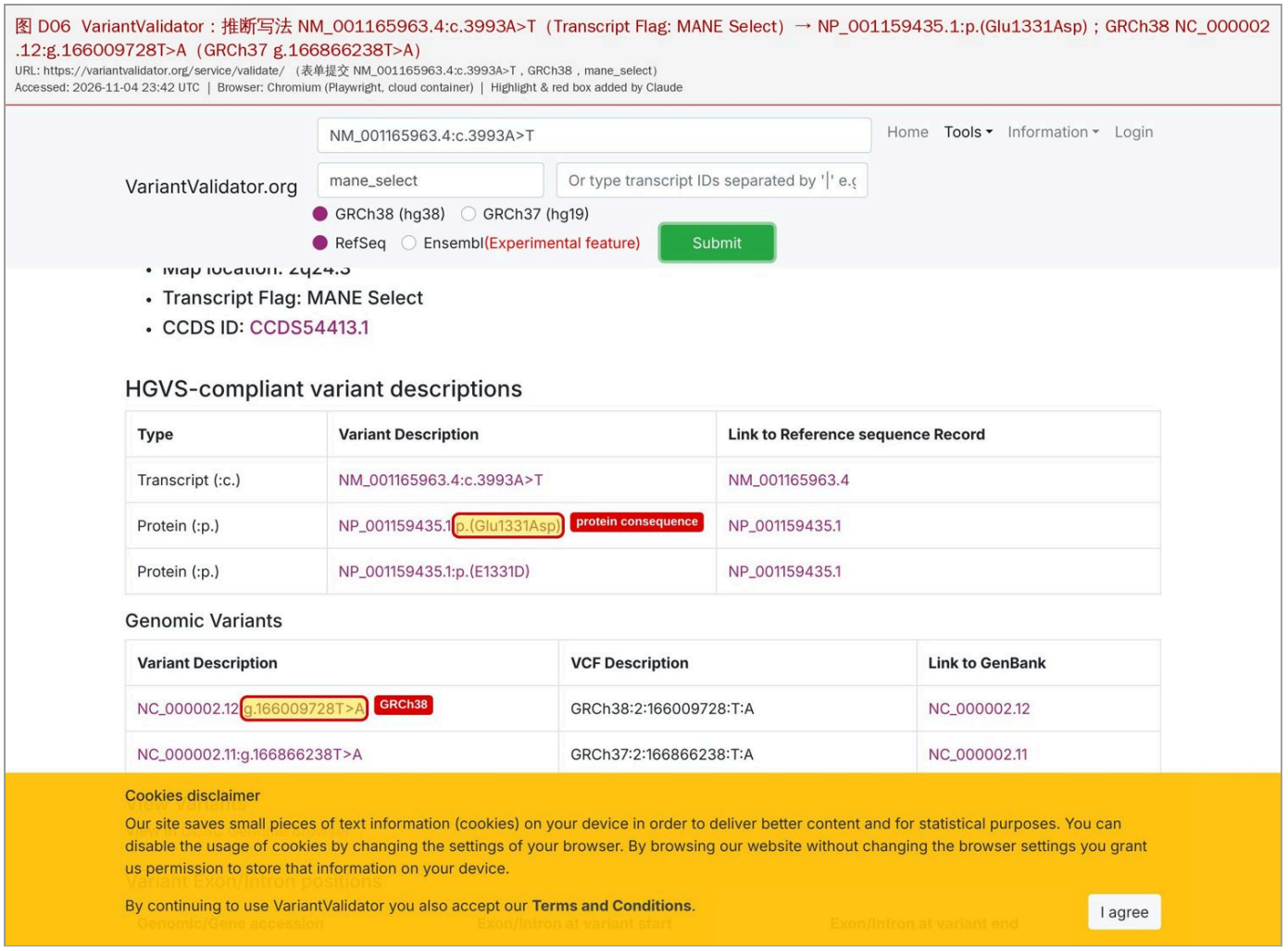


图 D06　VariantValidator：推断写法 NM_001165963.4:c.3993A>T（Transcript Flag: MANE Select）→ NP_001159435.1:p.(Glu1331Asp)；GRCh38 NC_000002.12:g.166009728T>A（GRCh37 g.166866238T>A）

文件：screenshots/D06_VariantValidator_c3993AT_Glu1331Asp_GRCh38.jpg ｜ 来源：https://variantvalidator.org/service/validate/ （表单提交 NM_001165963.4:c.3993A>T，GRCh38，mane_select） ｜ 访问：2026-11-04 23:42 UTC

A5　dbNSFP 注释：参考密码子 GAA、第 3 位、第 1331 位

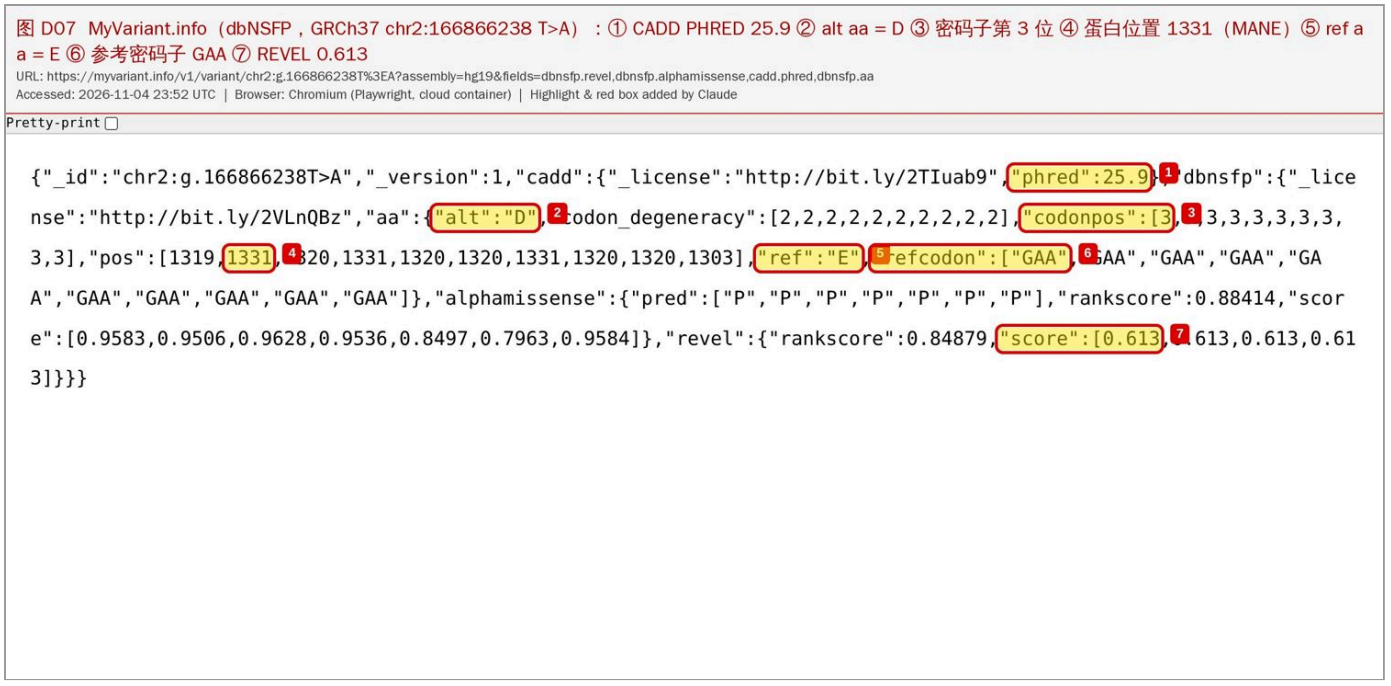


图 D07　MyVariant.info（dbNSFP，GRCh37 chr2:166866238 T>A）：① CADD PHRED 25.9 ② alt aa = D ③ 密码子第 3 位 ④ 蛋白位置 1331（MANE）⑤ ref aa = E ⑥ 参考密码子 GAA ⑦ REVEL 0.613

文件：screenshots/D07_dbNSFP_参考密码子GAA_REVEL.jpg ｜ 来源：https://myvariant.info/v1/variant/chr2:g.166866238T%3EA?assembly=hg19&fields=dbnsfp.revel,dbnsfp.alphamissense,cadd.phred,dbnsfp.aa ｜ 访问：2026-11-04 23:52 UTC

补充（脚本核对，未截图）：我把 NCBI 上 5 个 SCN1A RefSeq 转录本逐一翻译，c.3993 在全部转录本上都是 A；第 1331 位分别为 E（NM_001165963.4、NM_001202435.3、NM_001353948.2）、G（NM_006920.6）、L（NM_001165964.3），没有一个是 Asp。

B. 人群数据（gnomAD v4.1.2）

1. 打开区域 2-166009698-166009758（c.3993 前后各 30 bp），勾选 SNVs、Indels、Filtered variants、Exomes、Genomes（D08）。
2. 变异表按位置排序：2-166009725（c.3996）之后直接是 2-166009733（c.3988），2-166009728 没有任何变异。相邻位点 AN ≈ 1,605,000 个等位基因，远超 VCEP 要求的 10,000。
3. 覆盖度图（D09）：第 31 根柱 = chr2:166009728，约 0.95 的个体覆盖 ≥20×。gnomAD API 数值：外显子 mean 29.3×、98% ≥20×；基因组 mean 30.6×、94% ≥20×。
4. 直接打开两个可能变异的页面，均为 “Variant not found”（D10、D11）。gnomAD API 另外确认 2-166009728-T-C 也不存在（该位点三种替换都没有）。



图 D08　gnomAD v4.1.2 区域 2-166009698-166009758：勾选 Indels 与 Filtered variants 后共 26 个变异；2-166009725（p.Gly1332Gly）与 2-166009733（p.Phe1330Leu）之间没有 2-166009728 的任何变异；相邻位点 AN ≈ 1,605,000 个等位基因

文件：screenshots/D08_gnomAD_区域变异表_166009728无变异.jpg ｜ 来源：https://gnomad.broadinstitute.org/region/2-166009698-166009758?dataset=gnomad_r4 （已勾选 SNVs、Indels、Filtered variants、Exomes、Genomes） ｜ 访问：2026-11-04 23:46 UTC

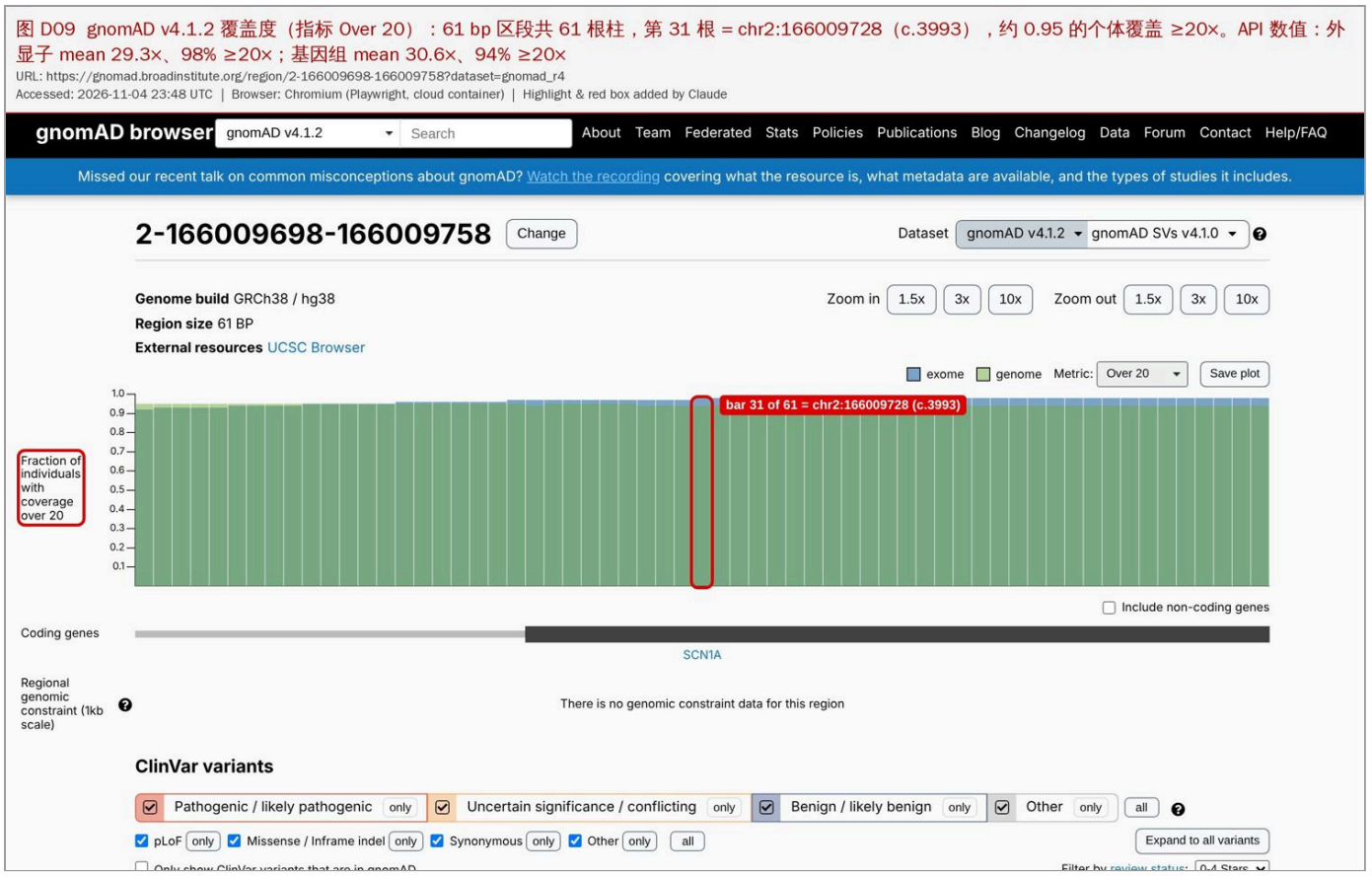


图 D09　gnomAD v4.1.2 覆盖度（指标 Over 20）：61 bp 区段共 61 根柱，第 31 根 = chr2:166009728（c.3993），约 0.95 的个体覆盖 ≥20×。API 数值：外显子 mean 29.3×、98% ≥20×；基因组 mean 30.6×、94% ≥20×

文件：screenshots/D09_gnomAD_覆盖度_166009728.jpg　|　来源：https://gnomad.broadinstitute.org/region/2-166009698-166009758?dataset=gnomad_r4　|　访问：2026-11-04 23:48 UTC

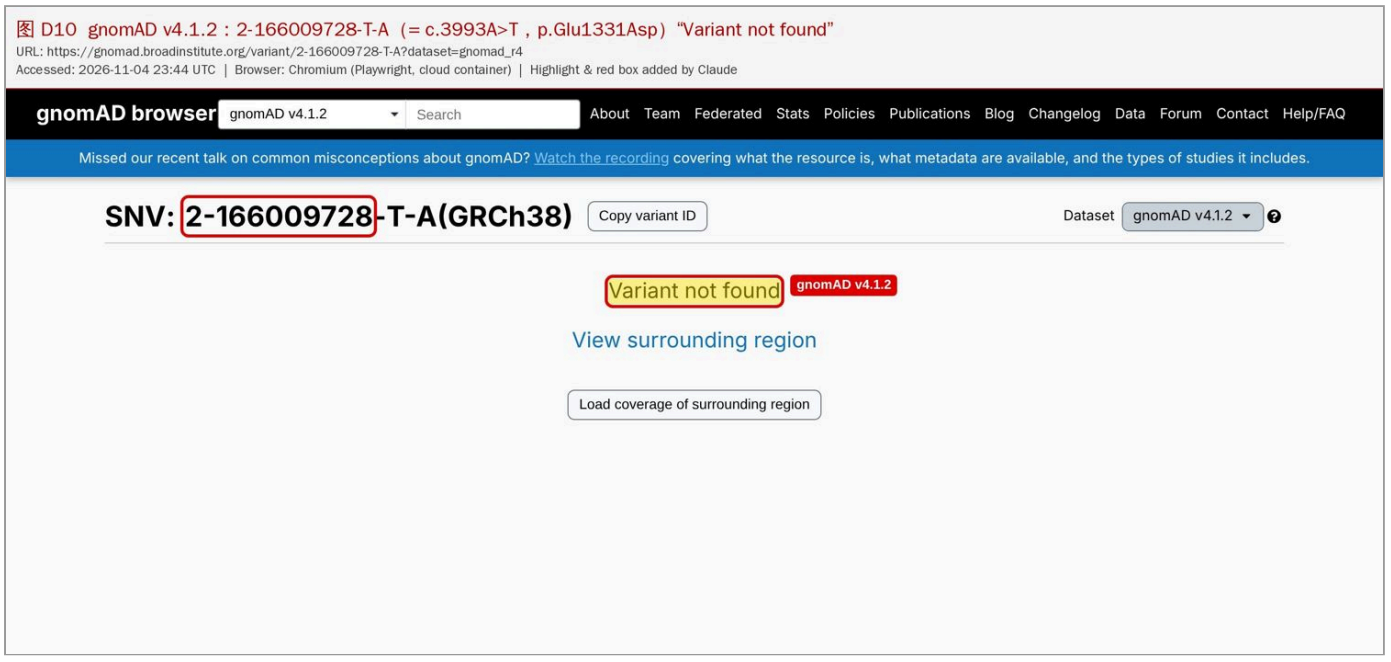


图 D10　gnomAD v4.1.2：2-166009728-T-A（= c.3993A>T，p.Glu1331Asp）“Variant not found”

文件：screenshots/D10_gnomAD_2-166009728-T-A_未找到.jpg　|　来源：https://gnomad.broadinstitute.org/variant/2-166009728-T-A?dataset=gnomad_r4　|　访问：2026-11-04 23:44 UTC

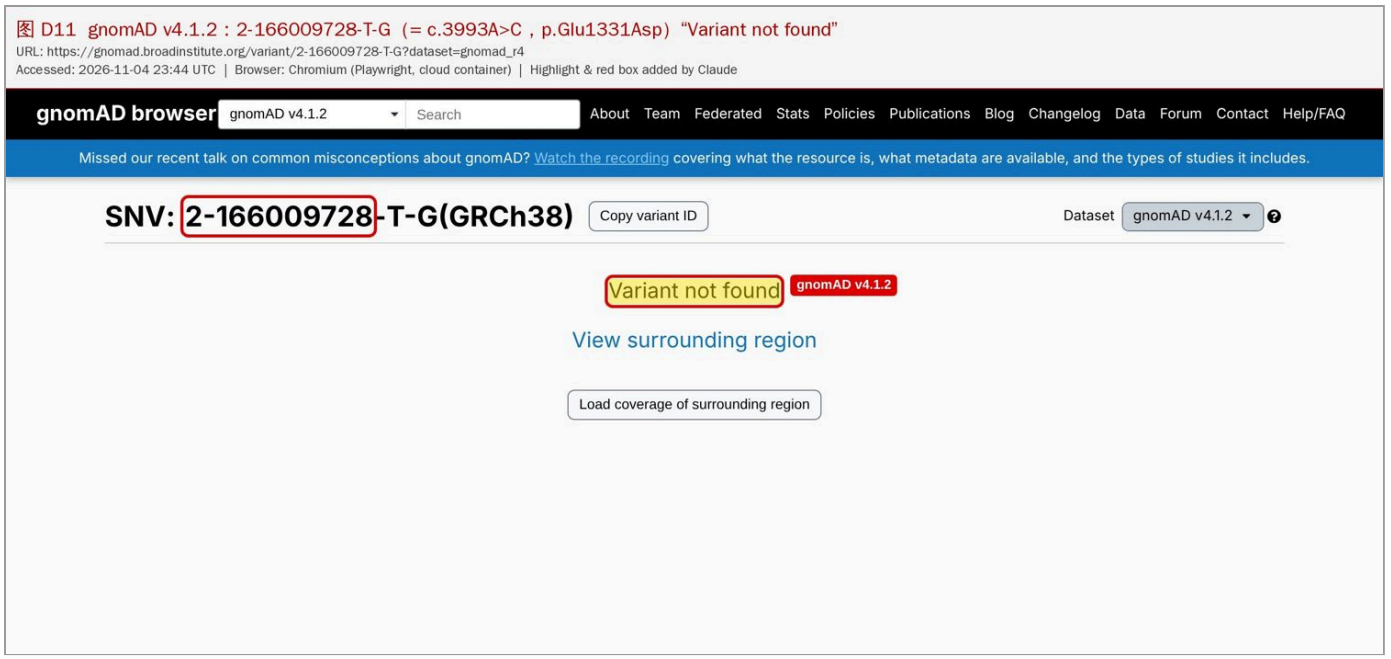


图 D11　gnomAD v4.1.2：2-166009728-T-G（= c.3993A>C，p.Glu1331Asp）“Variant not found”

文件：screenshots/D11_gnomAD_2-166009728-T-G_未找到.jpg ｜ 来源：https://gnomad.broadinstitute.org/variant/2-166009728-T-G?dataset=gnomad_r4 ｜ 访问：2026-11-04 23:44 UTC

C. 临床数据库

ClinVar 网页对容器浏览器弹验证码（未点），改用 NCBI E-utilities 在浏览器中显示检索结果。另用脚本下载了 SCN1A 全部 5,556 条 ClinVar 记录的摘要，按标题筛选第 1331 位：0 条；邻近 1328–1333 位的记录见 D13。



图 D12　ClinVar（NCBI E-utilities 检索；ClinVar 网页对容器浏览器弹验证码，未点）：与第 1331 号密码子（GRCh38 chr2:166009728–166009730）重叠的记录数 count = 0

文件：screenshots/D12_ClinVar_第1331位无记录.jpg ｜ 来源：https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=clinvar&term=2%5Bchr%5D+AND+166009728%3A166009730%5Bchrpos38%5D&retmode=json ｜ 访问：2026-11-04 23:49 UTC

图 D13 ClinVar（E-utilities esummary）：邻近残基的 P/LP 错义 —— p.Ser1328Pro（P/LP）、p.Arg1329Pro（P）、p.Arg1329Leu（P）、p.Met1333Ile（LP）

URL: https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esummary.fcgi?db=clinvar&id=2203183,2169505,1916226,2430161&retmode=json

Accessed: 2026-11-04 23:49 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

Pretty-print ☐

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图 D13　ClinVar（E-utilities esummary）：邻近残基的 P/LP 错义 —— p.Ser1328Pro（P/LP）、p.Arg1329Pro（P）、p.Arg1329Leu（P）、p.Met1333Ile（LP）

文件：screenshots/D13_ClinVar_邻近残基P_LP错义.jpg ｜ 来源：https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esummary.fcgi?db=clinvar&id=2203183,2169505,1916226,2430161&retmode=json ｜ 访问：2026-11-04 23:49 UTC

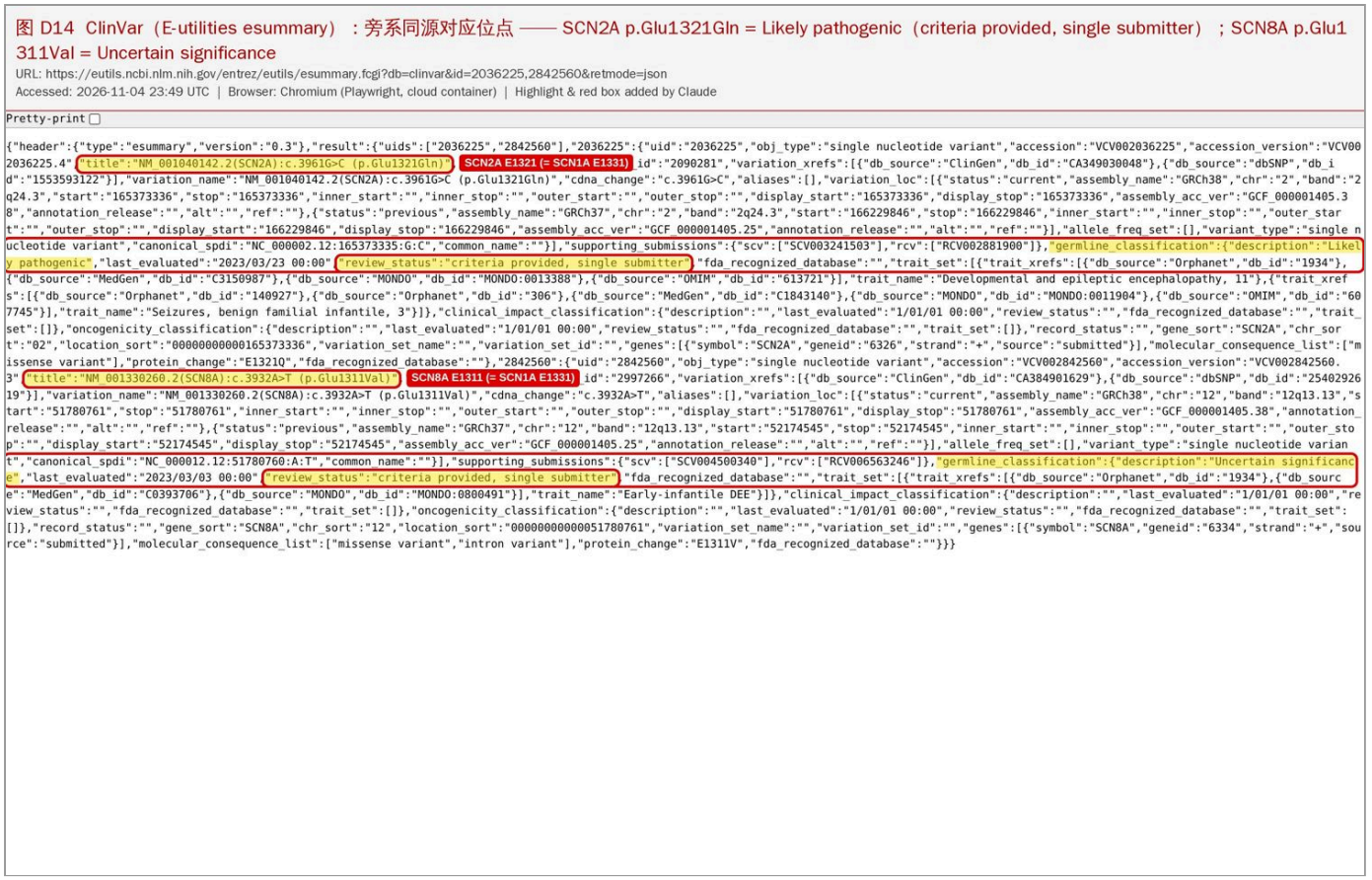


图 D14　ClinVar（E-utilities esummary）：旁系同源对应位点 —— SCN2A p.Glu1321Gln = Likely pathogenic（criteria provided, single submitter）；SCN8A p.Glu1311Val = Uncertain significance

文件：screenshots/D14_ClinVar_旁系同源对应位点.jpg ｜ 来源：https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esummary.fcgi?db=clinvar&id=2036225,2842560&retmode=json ｜ 访问：2026-11-04 23:49 UTC

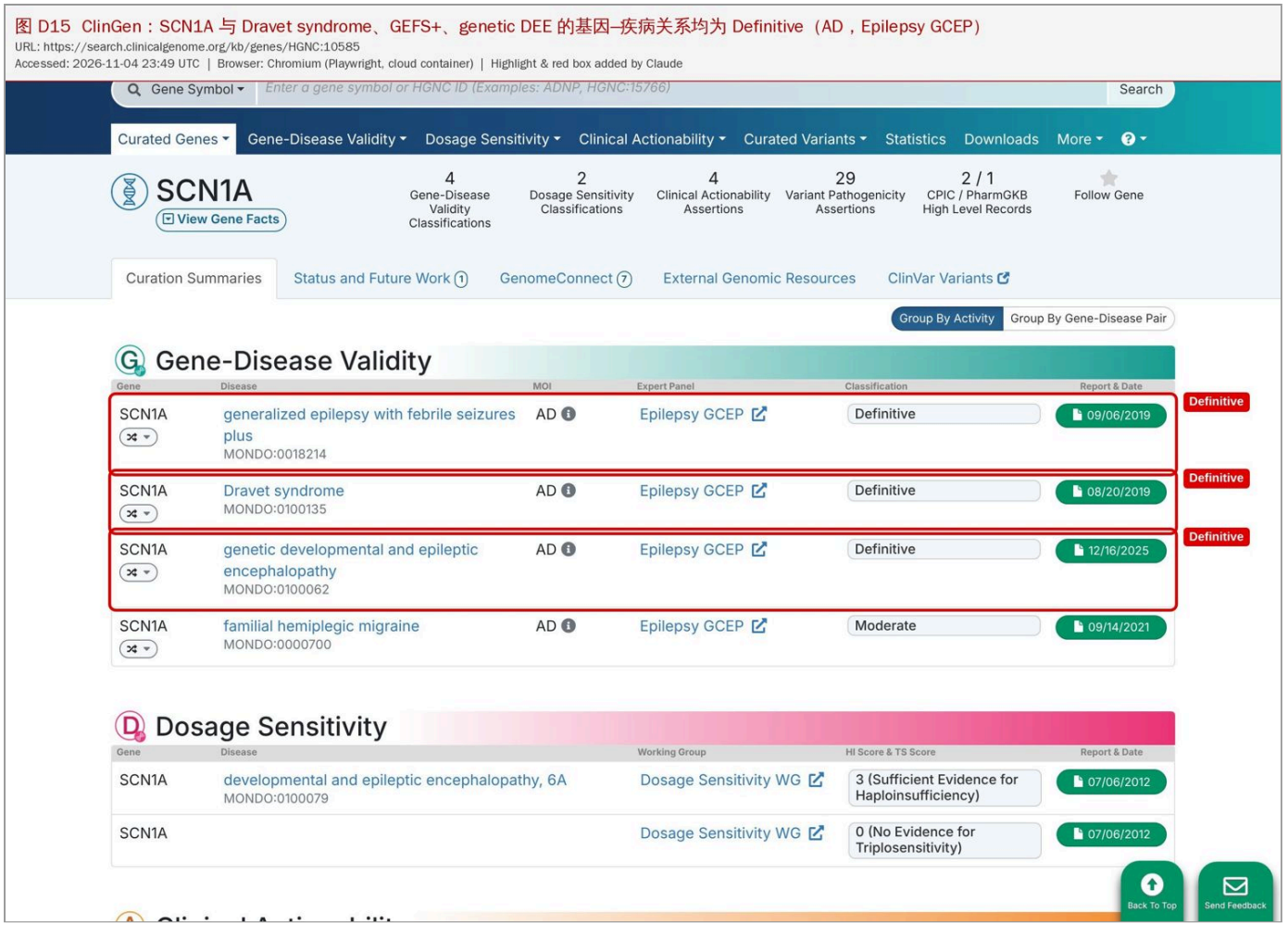


图 D15　ClinGen：SCN1A 与 Dravet syndrome、GEFS+、genetic DEE 的基因–疾病关系均为 Definitive（AD，Epilepsy GCEP）

文件：screenshots/D15_ClinGen_基因疾病有效性.jpg ｜ 来源：https://search.clinicalgenome.org/kb/genes/HGNC:10585 ｜ 访问：2026-11-04 23:49 UTC

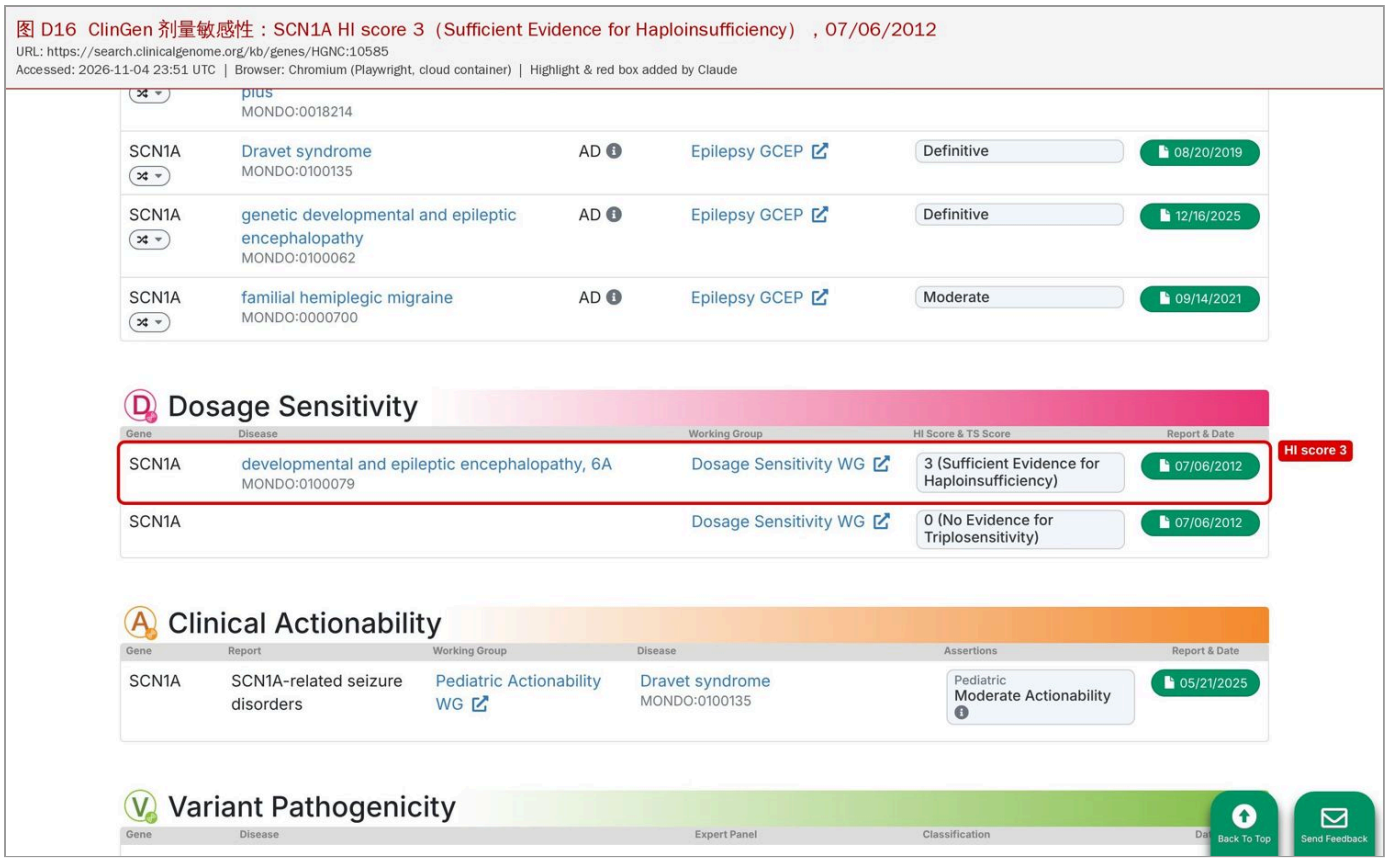


图 D16　ClinGen 剂量敏感性：SCN1A HI score 3（Sufficient Evidence for Haploinsufficiency），07/06/2012

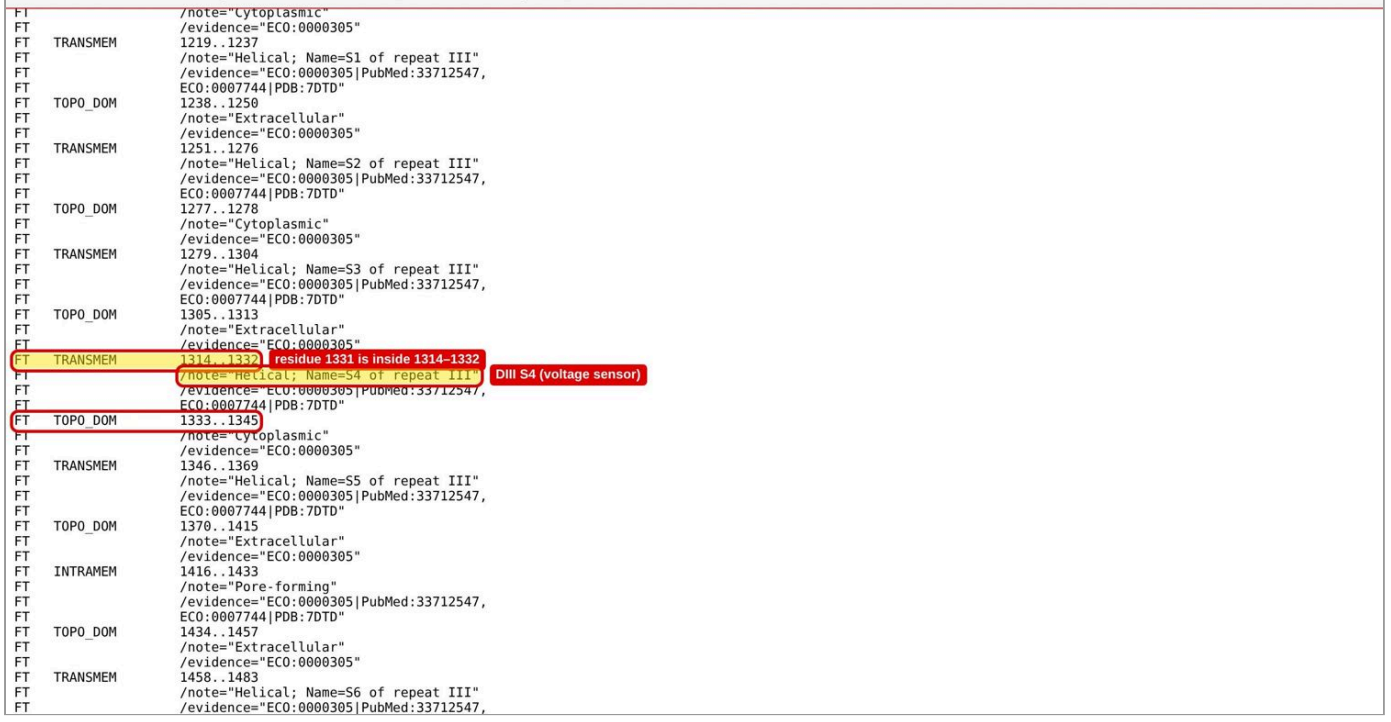
文件：screenshots/D16_ClinGen_剂量敏感性HI3.jpg ｜ 来源：https://search.clinicalgenome.org/kb/genes/HGNC:10585 ｜ 访问：2026-11-04 23:51 UTC

D. 蛋白位置

图 D17　UniProt P35498 条目（纯文本版）：FT TRANSMEM 1314..1332 /note="Helical; Name=S4 of repeat III"；1333..1345 为胞质侧（S4–S5 linker）。E1331 位于 DIII S4 电压感受器的胞内端

URL: https://rest.uniprot.org/uniprotkb/P35498.txt

Accessed: 2026-11-04 23:52 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude




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FT                   /evidence="ECO:0000305"
FT   TRANSMEM        1219..1237
FT                   /note="Helical; Name=S1 of repeat III"
FT                   /evidence="ECO:0000305|PubMed:33712547,
FT                   ECO:0007744|PDB:7DTD"
FT   TOPO_DOM        1238..1250
FT                   /note="Extracellular"
FT                   /evidence="ECO:0000305"
FT   TRANSMEM        1251..1276
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FT                   /evidence="ECO:0000305|PubMed:33712547,
FT                   ECO:0007744|PDB:7DTD"
FT   TOPO_DOM        1277..1278
FT                   /note="Cytoplasmic"
FT                   /evidence="ECO:0000305"
FT   TRANSMEM        1279..1304
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FT                   /evidence="ECO:0000305|PubMed:33712547,
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FT   TOPO_DOM        1305..1313
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FT                   ECO:0007744|PDB:7DTD"
FT   TOPO_DOM        1333..1345
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FT                   /evidence="ECO:0000305"
FT   TRANSMEM        1346..1369
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FT   TOPO_DOM        1370..1415
FT                   /note="Extracellular"
FT                   /evidence="ECO:0000305"
FT   INTRAMEM        1416..1433
FT                   /note="Pore-forming"
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FT                   ECO:0007744|PDB:7DTD"
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FT                   /evidence="ECO:0000305"
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图 D17　UniProt P35498 条目（纯文本版）：FT TRANSMEM 1314..1332 /note="Helical; Name=S4 of repeat III"；1333..1345 为胞质侧（S4–S5 linker）。E1331 位于 DIII S4 电压感受器的胞内端

文件：screenshots/D17_UniProt_拓扑_DIII_S4.jpg ｜ 来源：https://rest.uniprot.org/uniprotkb/P35498.txt ｜ 访问：2026-11-04 23:52 UTC

E. 计算预测

p.(Glu1331Asp) 可由 c.3993A>T 或 c.3993A>C 产生，蛋白层面完全相同；两者的 REVEL、AlphaMissense 一致。Ensembl VEP 接口查询时返回 503/500，故改用 dbNSFP（D07）和 SpliceAI-lookup（D18–D21）。

REVEL 阈值（Pejaver 等 2022，表 2；VCEP 指定 REVEL）：PP3_Supporting ≥0.644，Moderate ≥0.773，Strong ≥0.932；BP4_Supporting ≤0.290，Moderate ≤0.183，Strong ≤0.016，Very Strong ≤0.003。0.613 落在 0.290–0.644 之间 → PP3 和 BP4 都不适用。阈值是通过 PMC 开放获取全文接口（PMC9748256）读出来核对的；PMC 网页对容器浏览器弹验证码，【需查】原文表 2 截图请在你的 Chrome 补。SpliceAI-lookup 页面给 REVEL 0.613 记 0 分，与此一致（D19）。

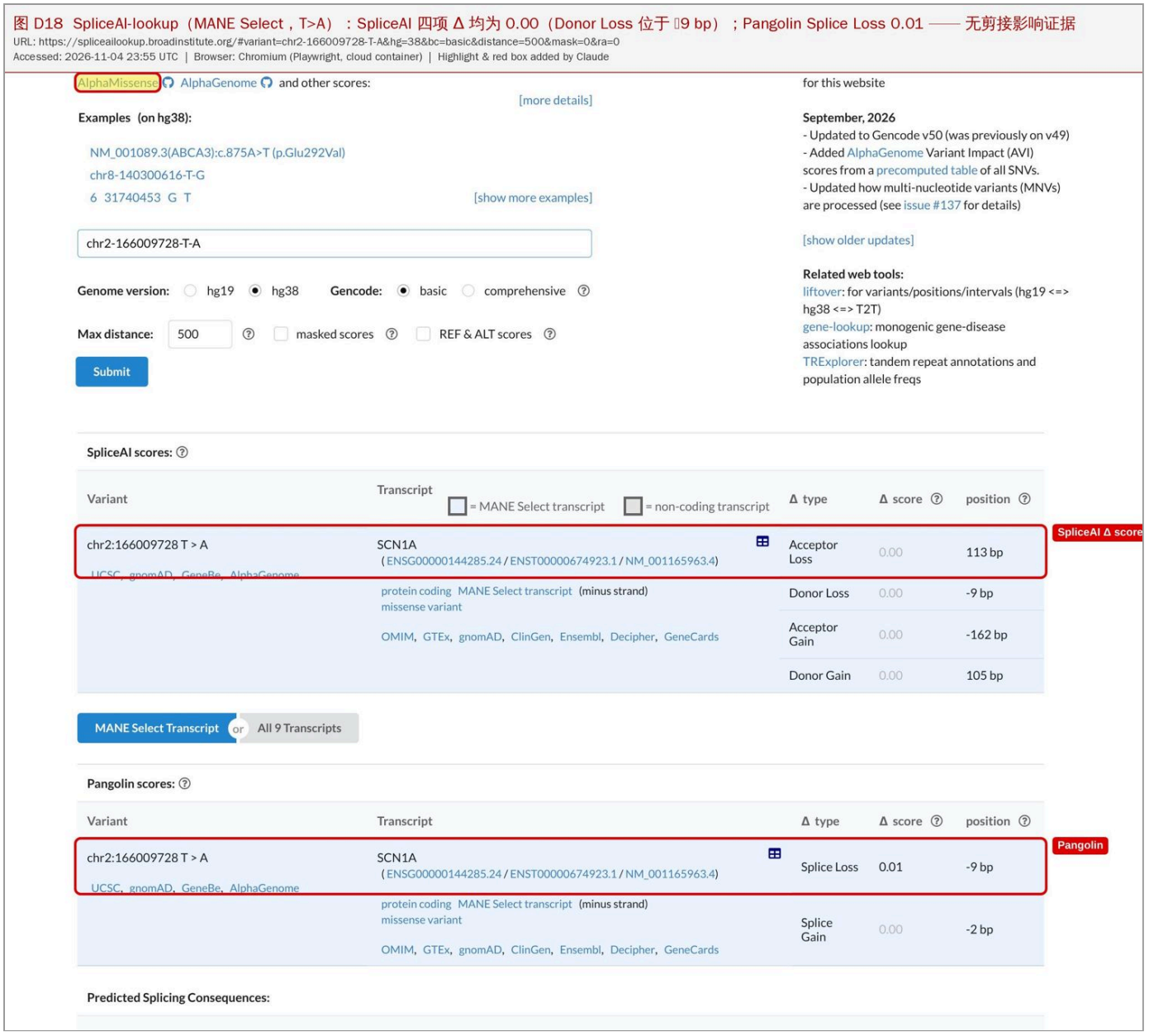


图 D18　SpliceAI-lookup（MANE Select，T>A）：SpliceAI 四项 Δ 均为 0.00（Donor Loss 位于 −9 bp）；Pangolin Splice Loss 0.01 —— 无剪接影响证据

文件：screenshots/D18_SpliceAI_TA_剪接预测.jpg ｜ 来源：https://spliceailookup.broadinstitute.org/#variant=chr2-166009728-T-A&hg=38&bc=basic&distance=500&mask=0&ra=0 ｜ 访问：2026-11-04 23:55 UTC

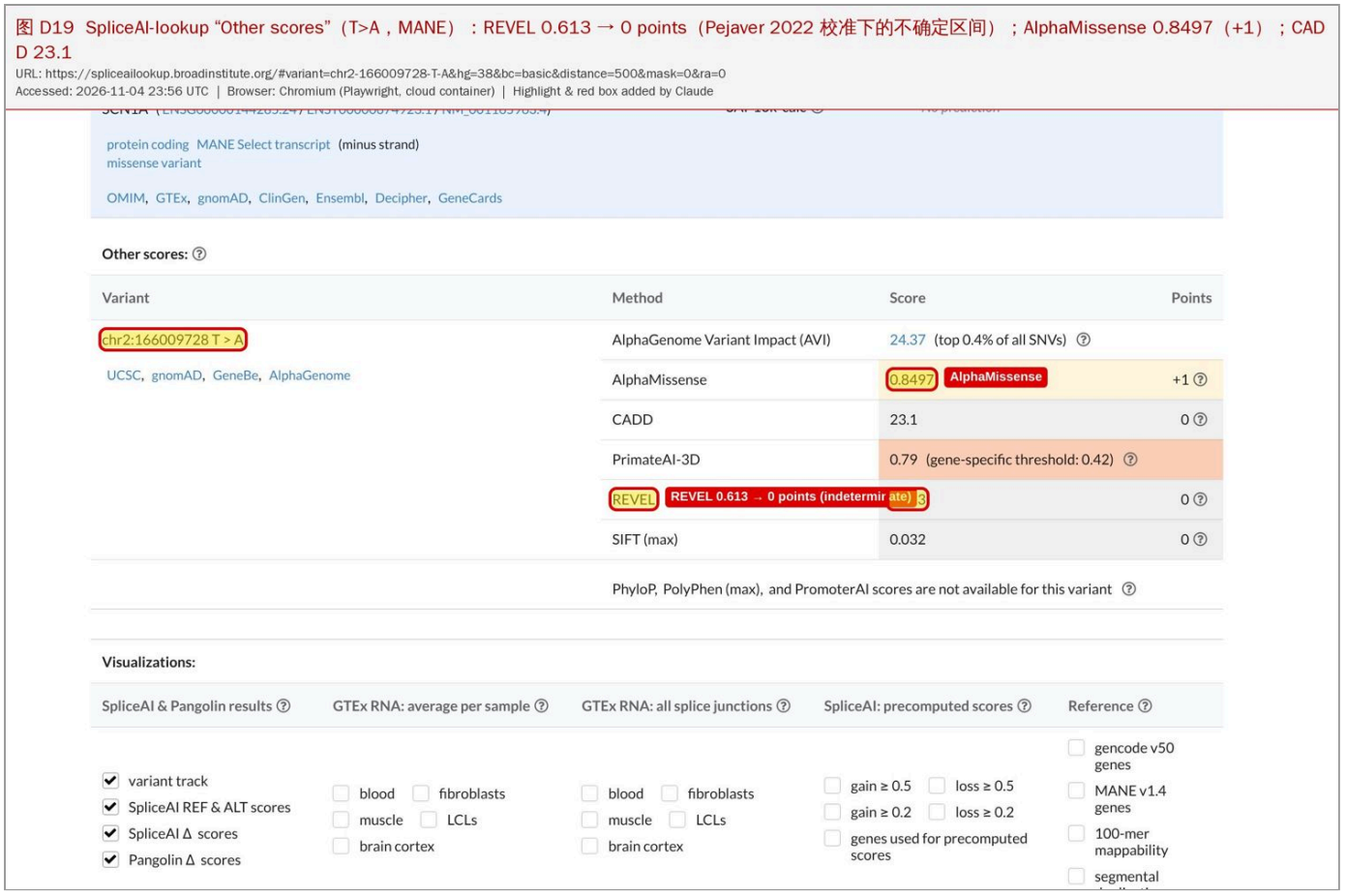


图 D19　SpliceAI-lookup “Other scores”（T>A，MANE）：REVEL 0.613 → 0 points（Pejaver 2022 校准下的不确定区间）；AlphaMissense 0.8497（+1）；CADD 23.1

文件：screenshots/D19_SpliceAI_TA_REVEL_AlphaMissense.jpg ｜ 来源：https://spliceailookup.broadinstitute.org/#variant=chr2-166009728-T-A&hg=38&bc=basic&distance=500&mask=0&ra=0 ｜ 访问：2026-11-04 23:56 UTC

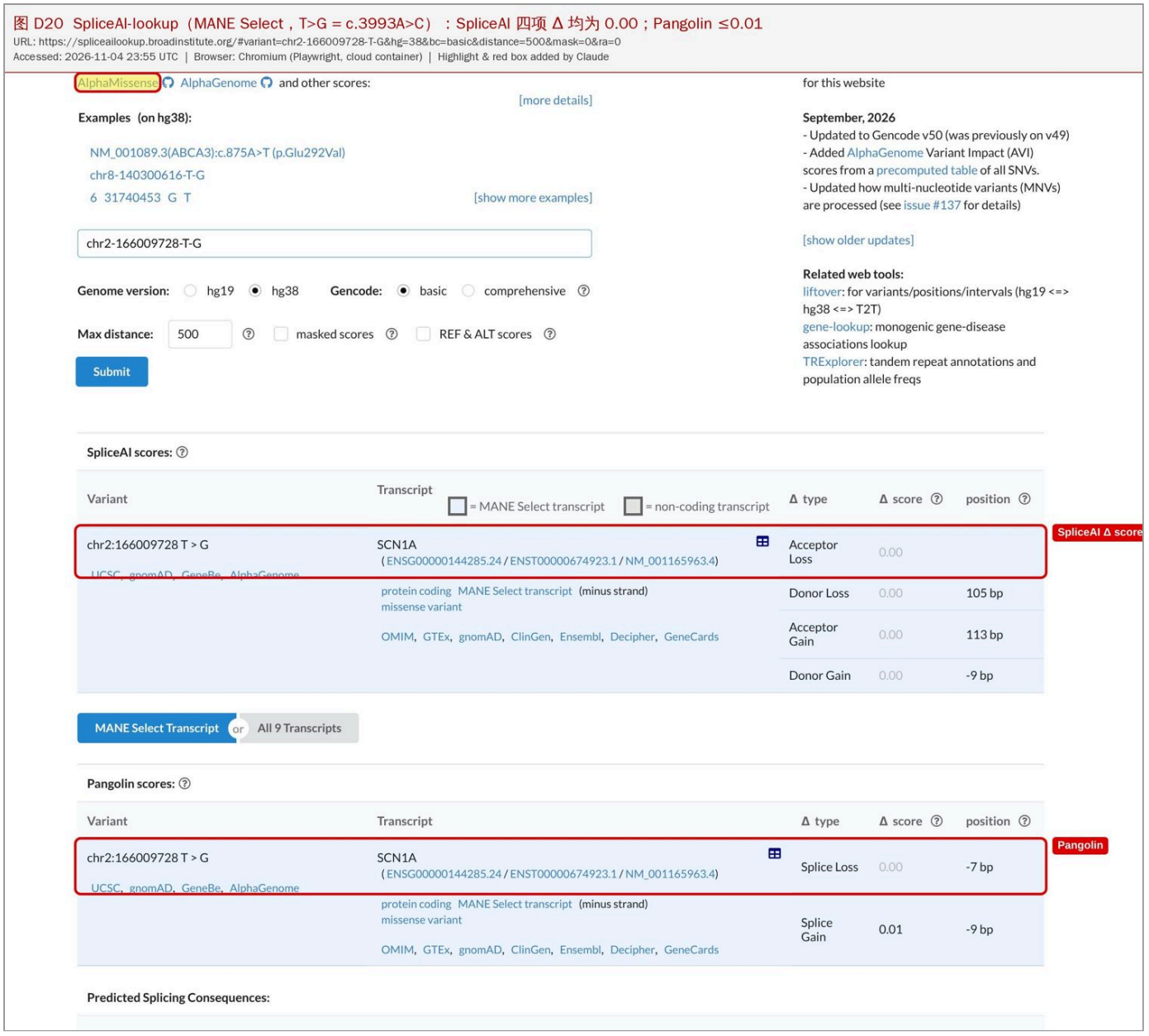


图 D20　SpliceAI-lookup（MANE Select，T>G = c.3993A>C）：SpliceAI 四项 Δ 均为 0.00；Pangolin ≤0.01

文件：screenshots/D20_SpliceAI_TG_剪接预测.jpg ｜ 来源：https://spliceailookup.broadinstitute.org/#variant=chr2-166009728-T-G&hg=38&bc=basic&distance=500&mask=0&ra=0 ｜ 访问：2026-11-04 23:55 UTC

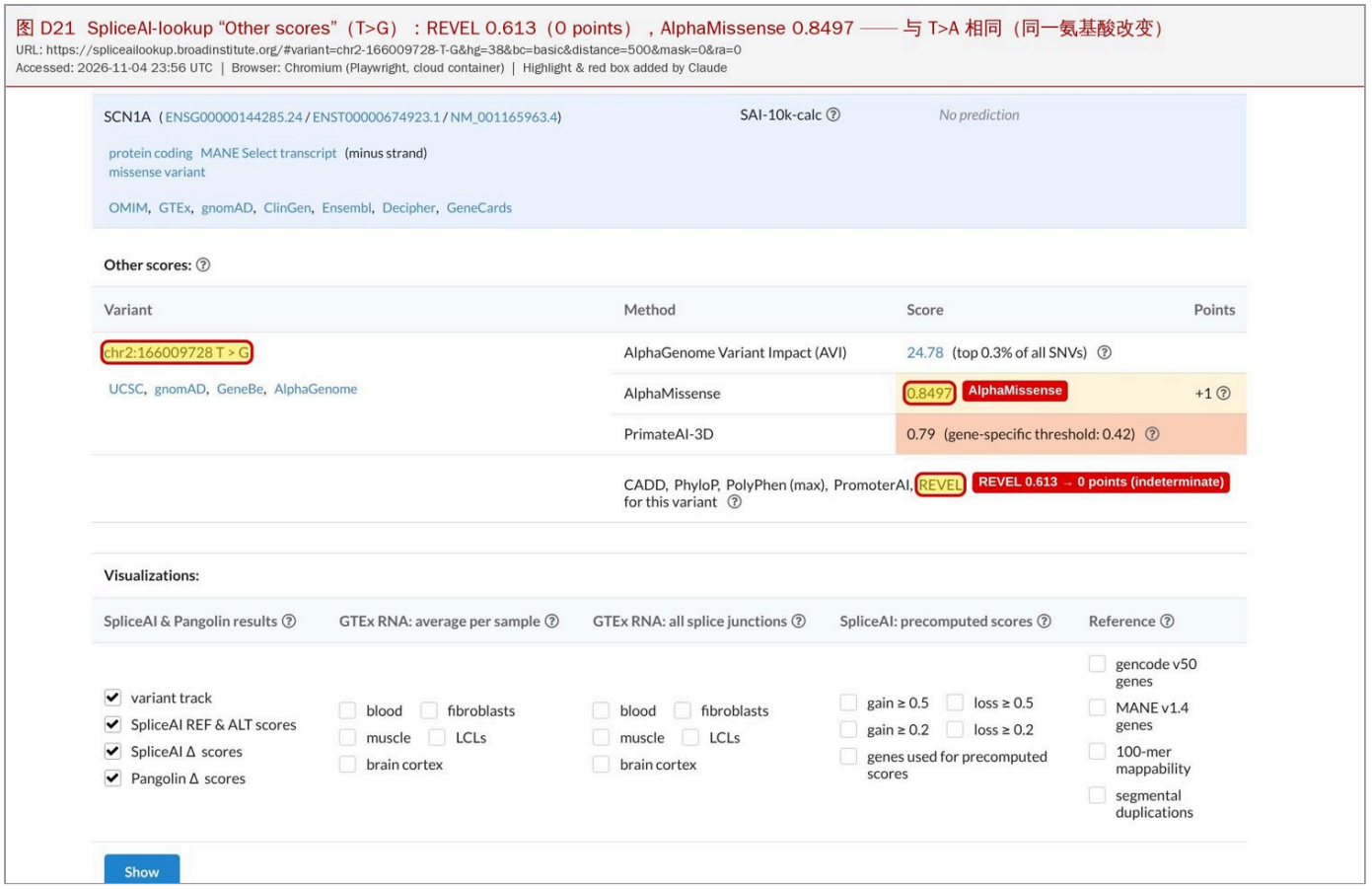


图 D21　SpliceAI-lookup “Other scores”（T>G）：REVEL 0.613（0 points），AlphaMissense 0.8497 —— 与 T>A 相同（同一氨基酸改变）

文件：screenshots/D21_SpliceAI_TG_REVEL_AlphaMissense.jpg ｜ 来源：https://spliceailookup.broadinstitute.org/#variant=chr2-166009728-T-G&hg=38&bc=basic&distance=500&mask=0&ra=0 ｜ 访问：2026-11-04 23:56 UTC

保守性：手动比对（表 A1）

方法：从 UniProt REST 取序列，以人 SCN1A 1311–1351 为查询序列，用 Smith–Waterman 局部比对（BLOSUM62，线性空位罚分 −4，我自己写的脚本）定位对应残基。这是手动比对，不是数据库现成结果；UCSC 比对/phyloP 页面查询时超时，未能截图（dbNSFP 给出的 phyloP100way = 0.615，对应摆动位的核苷酸保守性，意义有限）。黄色列 = 人 SCN1A 第 1331 位。

物种 / 基因	UniProt	比对窗口（人 1311–1351）	对应残基	位置
Human SCN1A	P35498 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1331
Mouse Scn1a	A2APX8 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1331
Rat Scn1a	P04774 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1331
Cow SCN1A	M0QW12 (TrEMBL)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1331
Chicken SCN1A	A0A8V1A8W9 (TrEMBL)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1338
Xenopus tropicalis scn1a	A0A8J1IT78 (TrEMBL)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALIGAIPSIMNVL	E	1329
Zebrafish scn1lab	F1QXF5 (TrEMBL)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1312
Zebrafish scn1laa	Q20JQ9 (TrEMBL)	AIKSLRTLRAFRPLRALSRFD**D**GMRVVVNALLGAIPSIMNVL	D	1280
Drosophila para	P35500 (Swiss-Prot)	AFKTMRTLRALRPLRAMSRMQ**Q**GMRVVVNALVQAIPSIFNVL	Q	1414
Human SCN2A	Q99250 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALLGAIPSIMNVL	E	1321
Human SCN3A	Q9NY46 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFE**E**GMRVVVNALVGAIPSIMNVL	E	1319

Human SCN8A	Q9UQD0 (Swiss-Prot)	AIKSLRTLRALRPLRALSRFEGMRVVVNALVGAIPSIMNVL	E	1311
Human SCN4A	P35499 (Swiss-Prot)	-IKSLRTLRALRPLRALSRFEGMRVVVNALLGAIPSIMNVL	E	1144
Human SCN5A	Q14524 (Swiss-Prot)	-IKSLRTLRALRPLRALSRFEGMRVVVNALVGAIPSIMNVL	E	1318
Human SCN9A	Q15858 (Swiss-Prot)	-IKSLRTLRALRPLRALSRFEGMRVVVNALIGAIPSIMNVL	E	1305
Human SCN10A	Q9Y5Y9 (Swiss-Prot)	-IKALRTLRALRPLRALSRFEGMRVVVDALVGAIPSIMNVL	E	1265
Human SCN11A	Q9UI33 (Swiss-Prot)	-LKSFRTLRALRPLRALSQFEGMKVVVNALIGAIPAILNVL	E	1162

解读：Glu 在全部 9 个人类电压门控钠通道 α 亚基、以及哺乳动物、鸡、爪蟾、斑马鱼 scn1lab 中保守，支持该位点功能重要；但斑马鱼 scn1laa 天然为 Asp，说明 E→D 这一保守替换至少在一个脊椎动物同源通道中可以耐受。这一点与 REVEL 处于不确定区间相符，写进报告时用 suggests / is consistent with。

F. ClinGen Epilepsy Sodium Channel VCEP《SCN1A 细则》v2.0（GN067）

附件 PM1 Table、Paralogous Gene Table、PVS1 exon numbering 为 xlsx，在页面 Files & Images 点击下载按钮获得（保存在 ClinGen_VCEP附件/），D28、D29 是本地渲染后的截图，数值未改动。附件 “Combining Rules”（VCEP 的积分合并规则）和 “PS1_Variants impacting splicing” 是 PDF，按“PDF 只在你要求时下载”的规则没有下载。

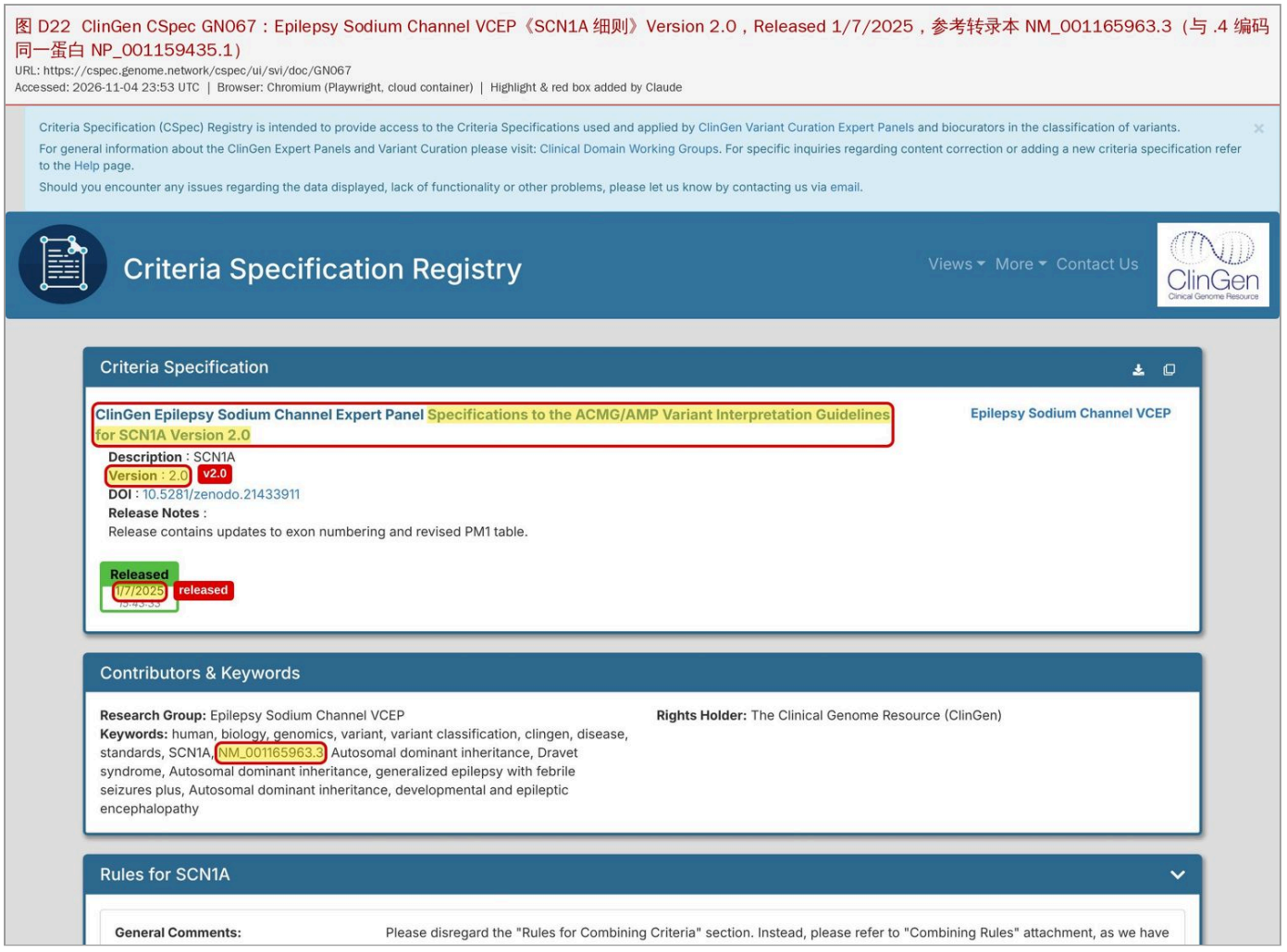


图 D22　ClinGen CSpec GN067：Epilepsy Sodium Channel VCEP《SCN1A 细则》Version 2.0，Released 1/7/2025，参考转录本 NM_001165963.3（与 .4 编码同一蛋白 NP_001159435.1）

文件：screenshots/D22_ClinGen_VCEP_SCN1A细则v2.0.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-04 23:53 UTC

图 D23 VCEP 细则：PM1 = Moderate（位于 Pathogenic Enriched Region，见附件 PM1 Table）；PM2 只能用 Supporting（≥10,000 个等位基因中 ≤1 个）
URL: https://cspec.genome.network/cspec/ui/svi/doc/GN067
Accessed: 2026-11-04 23:53 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

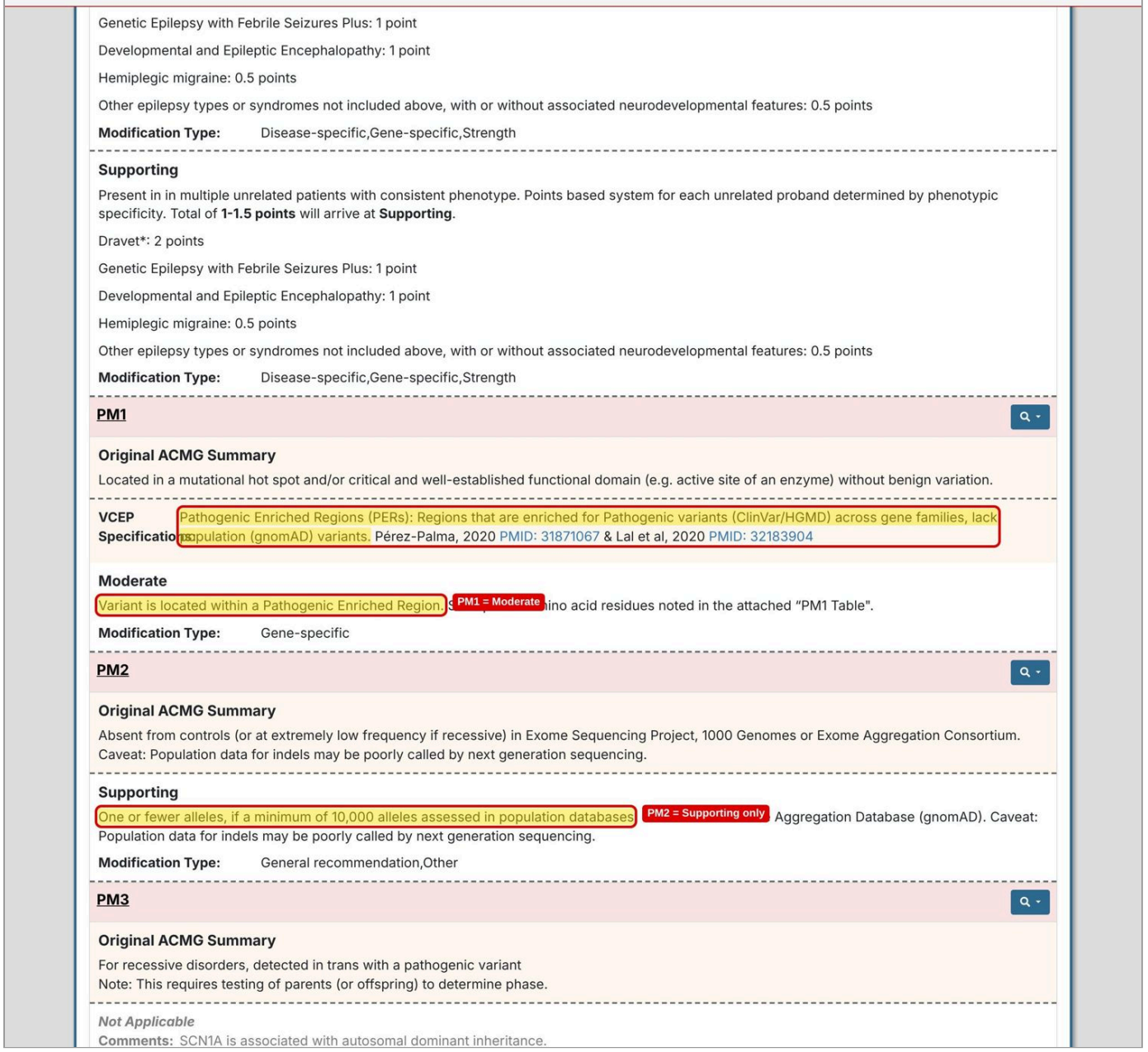


图 D23　VCEP 细则：PM1 = Moderate（位于 Pathogenic Enriched Region，见附件 PM1 Table）；PM2 只能用 Supporting（≥10,000 个等位基因中 ≤1 个）

文件：screenshots/D23_VCEP_PM1_PM2.jpg　|　来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067　|　访问：2026-11-04 23:53 UTC

图 D24 VCEP 细则：PP2 不适用（良性错义常见）；PP3 用 REVEL 按 Pejaver 2022（PMID 36413997）判定，最高 Moderate，PP3+PM1 合计不超过 Strong；PP4 不适用（表型特异性已并入 PS2/PM6/PS4）
URL: https://cspec.genome.network/cspec/ui/svi/doc/GN067
Accessed: 2026-11-04 23:53 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

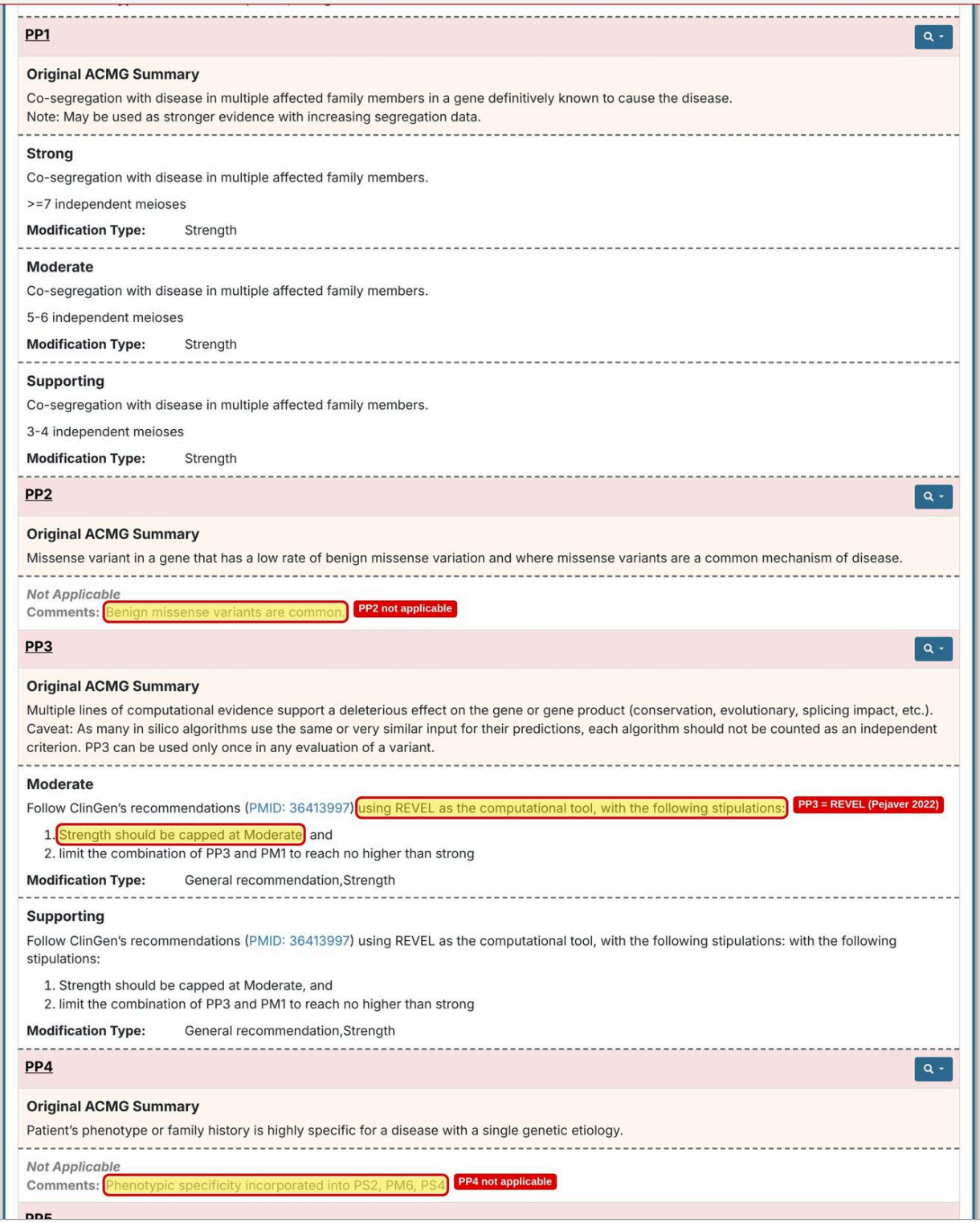


图 D24　VCEP 细则：PP2 不适用（良性错义常见）；PP3 用 REVEL 按 Pejaver 2022（PMID 36413997）判定，最高 Moderate，PP3+PM1 合计不超过 Strong；PP4 不适用（表型特异性已并入 PS2/PM6/PS4）

文件：screenshots/D24_VCEP_PP2_PP3_PP4.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-04 23:53 UTC

图 D25 VCEP 细则 PS2：Dravet 临床标准（Li 2021，PMID 34338318；含 Seizure onset ≤12m 等）；确认新发且表型为 Dravet 计 2 分 = PS2 Strong，GEFS+ 计 1 分

URL: https://cspec.genome.network/cspec/ui/svi/doc/GN067
Accessed: 2026-11-04 23:54 UTC | Browser: Chromium (Playwright, cloud container) | Highlight & red box added by Claude

+/- 1,2 dinucleotide position). Specific combinations are outlined in Table 2 in Walker, et al (2023) PMID: 37352859, also provided as a supplement ("PS1_Variants impacting splicing").

Modification Type: Gene-specific,Strength

Supporting

Same/identical amino acid change as previously reported (Caveat: beware of changes that impact splicing rather than at the amino acid/protein level).

- A single identical amino acid change in a paralogous gene previously established as **Pathogenic or Likely Pathogenic**, including NDD genes with equivalent constraint scores (SCN1A, SCN2A, SCN3A, SCN8A). See Paralogous Gene Table for corresponding amino acid positions.

Same predicted impact on splicing as previously classified variant (Refer to Table 2 in Walker et al, 2023).

- PS1 can be applied at varying strengths for splice variants, in conjunction with either PP3 or PVS1. PS1 strength depends on location of the variant under assessment (within or outside the +/- 1,2 dinucleotide positions) and the location of the previously classified variant (within or outside the +/- 1,2 dinucleotide position). Specific combinations are outlined in Table 2 in Walker, et al (2023) PMID: 37352859, also provided as a supplement ("PS1_Variants impacting splicing").

Modification Type: Gene-specific

PS2

Original ACMG Summary

De novo (both maternity and paternity confirmed) in a patient with the disease and no family history.
Note: Confirmation of paternity only is insufficient. Egg donation, surrogate motherhood, errors in embryo transfer, etc. can contribute to non-maternity.

VCEP Specifications: Clinical Criteria for Dravet Syndrome: Based on Li et al, 2021 (PMID: 34338318) — Dravet criteria (for points)
Major features
-Fever-sensitive seizures,
-Hemiclonic or tonic-clonic seizures,
-Status epilepticus
Minor criteria (at least 3)
-Normal development prior to seizure onset
-Seizure onset <=12m
-Developmental impairment by age 5y
-Intellectual disability

Very Strong

De novo (both maternity and paternity confirmed) in a patient with the disease and no family history. Points based system for each unrelated proband determined by phenotypic specificity. Total of **4 points** will arrive at **Very Strong**.

Dravet*: 2 points

Genetic Epilepsy with Febrile Seizures Plus: 1 point

Developmental and Epileptic Encephalopathy: 1 point

Hemiplegic migraine: 0.5 points

Other epilepsy types or syndromes not included above, with or without associated neurodevelopmental features: 0.5 points

Note: Confirmation of paternity only is insufficient. Egg donation, surrogate motherhood, errors in embryo transfer, etc. can contribute to non-maternity.

Modification Type: Disease-specific,Strength

Strong

De novo (both maternity and paternity confirmed) in a patient with the disease and no family history. Points based system for each unrelated proband determined by phenotypic specificity. Total of 2 points will arrive at **Strong** — PS2 Strong = 2 points

Dravet*: 2 points

Genetic Epilepsy with Febrile Seizures Plus: 1 point

Developmental and Epileptic Encephalopathy: 1 point

Hemiplegic migraine: 0.5 points

Other epilepsy types or syndromes not included above, with or without associated neurodevelopmental features: 0.5 points

Note: Confirmation of paternity only is insufficient. Egg donation, surrogate motherhood, errors in embryo transfer, etc. can contribute to non-maternity.

Modification Type: Disease-specific,Strength

Moderate

De novo (both maternity and paternity confirmed) in a patient with the disease and no family history. Points based system for each unrelated proband determined by phenotypic specificity. Total of **1 point** will arrive at **Moderate**.

Dravet*: 2 points

Genetic Epilepsy with Febrile Seizures Plus: 1 point

Developmental and Epileptic Encephalopathy: 1 point

Hemiplegic migraine: 0.5 points

图 D25 VCEP 细则 PS2：Dravet 临床标准（Li 2021，PMID 34338318；含 Seizure onset ≤12m 等）；确认新发且表型为 Dravet 计 2 分 = PS2 Strong，GEFS+ 计 1 分

文件：screenshots/D25_VCEP_PS2_Dravet标准与积分.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-04 23:54 UTC

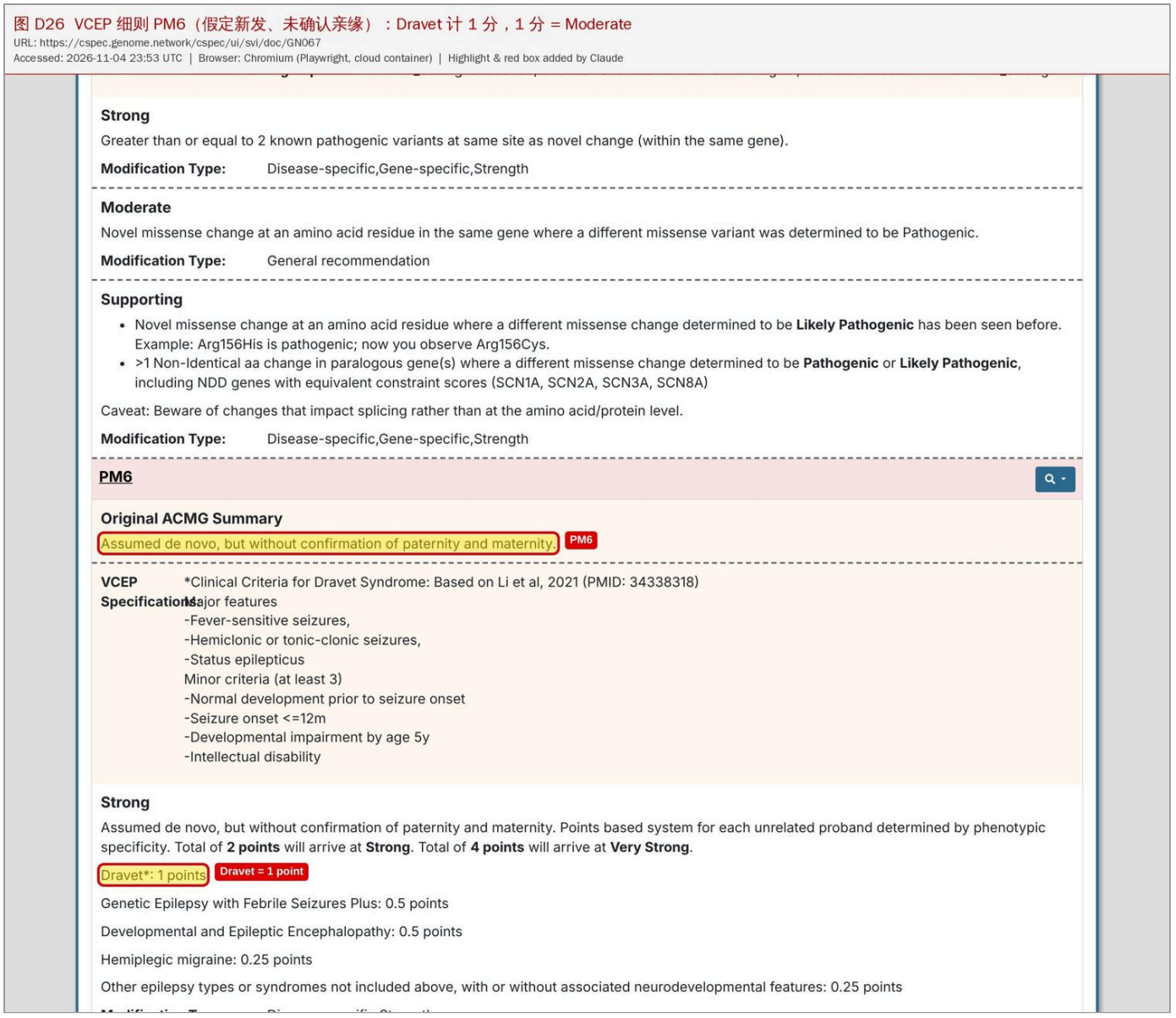


图 D26　VCEP 细则 PM6（假定新发、未确认亲缘）：Dravet 计 1 分，1 分 = Moderate

文件：screenshots/D26_VCEP_PM6积分.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-04 23:53 UTC

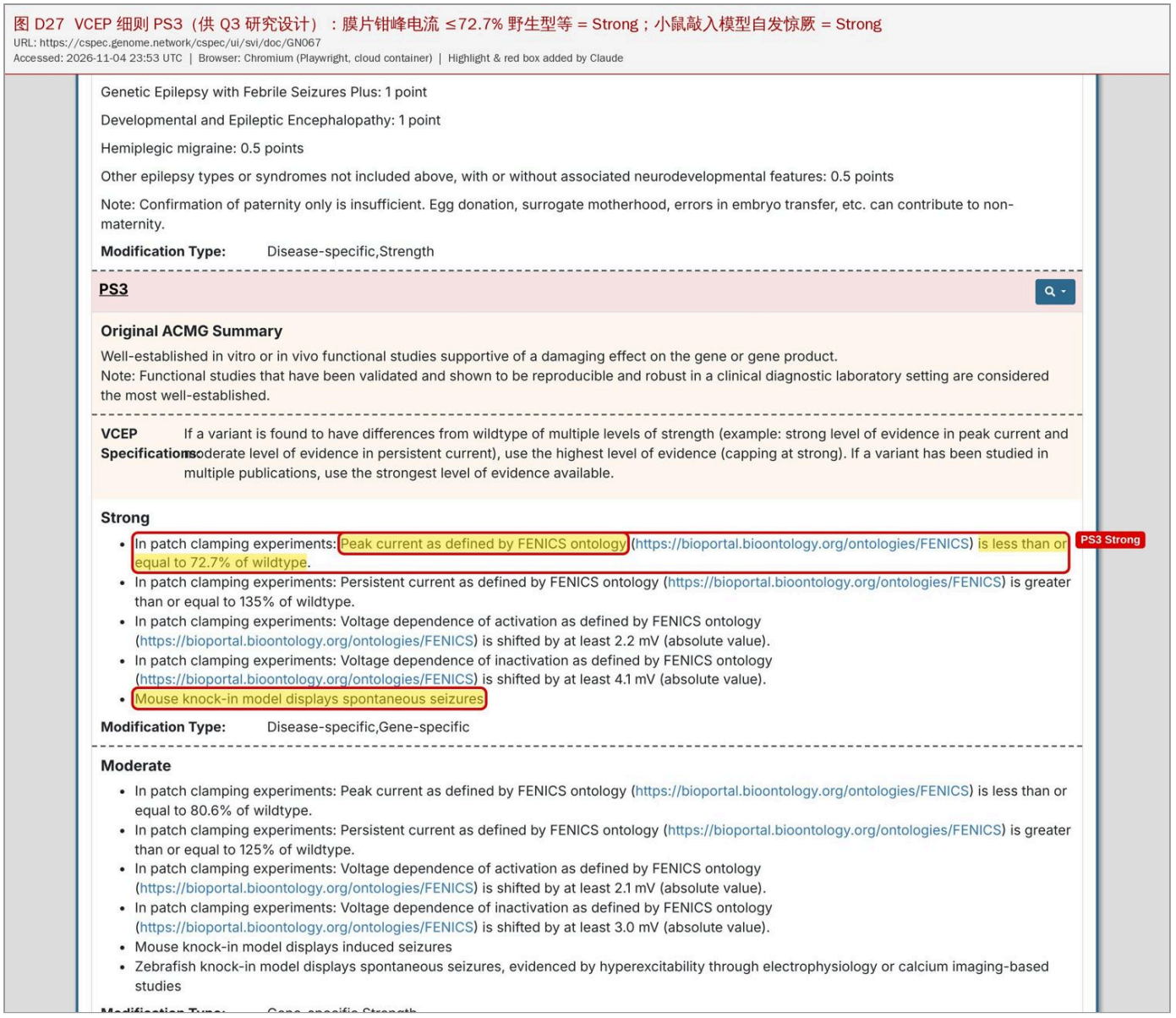
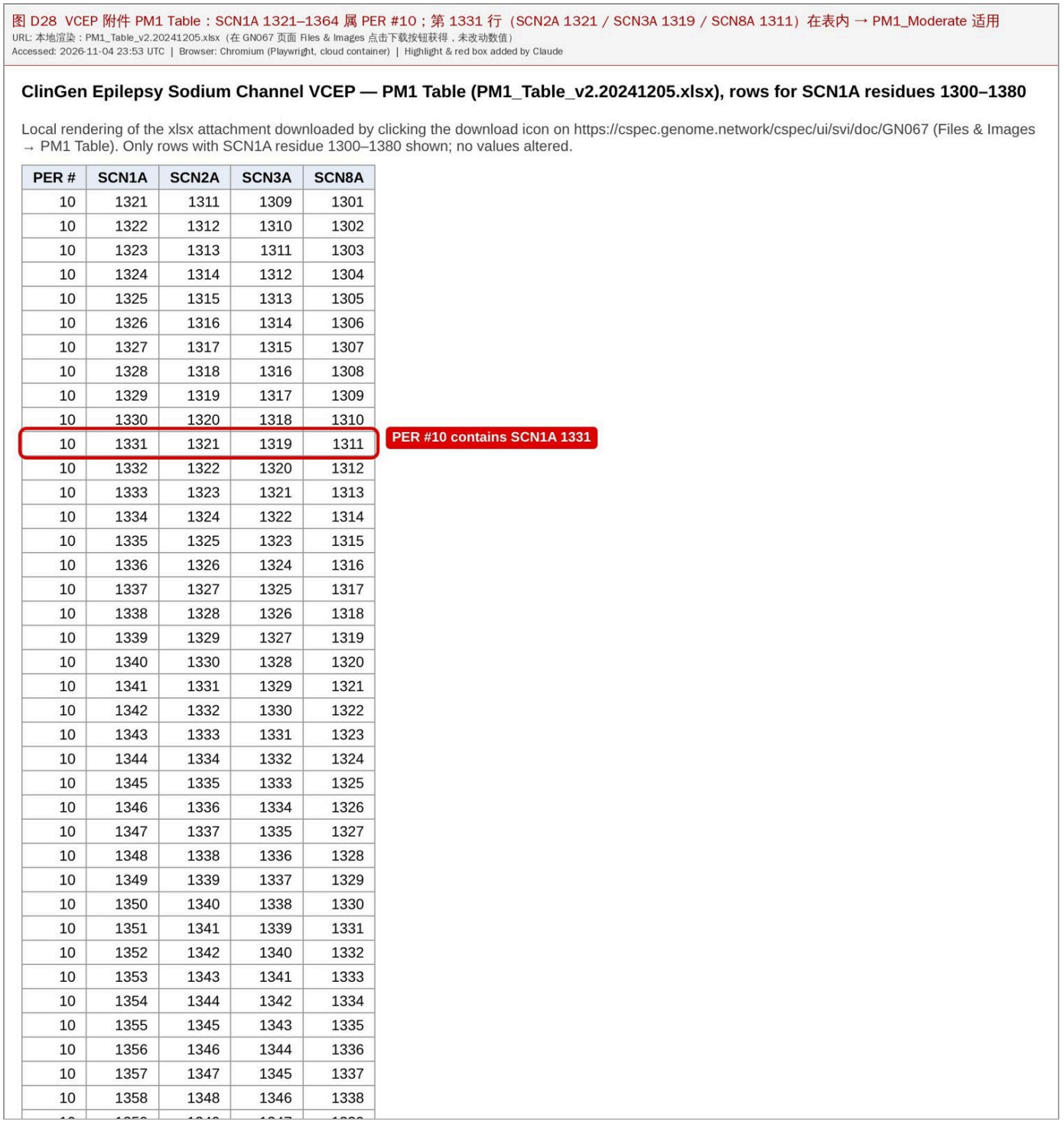


图 D27　VCEP 细则 PS3（供 Q3 研究设计）：膜片钳峰电流 ≤72.7% 野生型等 = Strong；小鼠敲入模型自发惊厥 = Strong

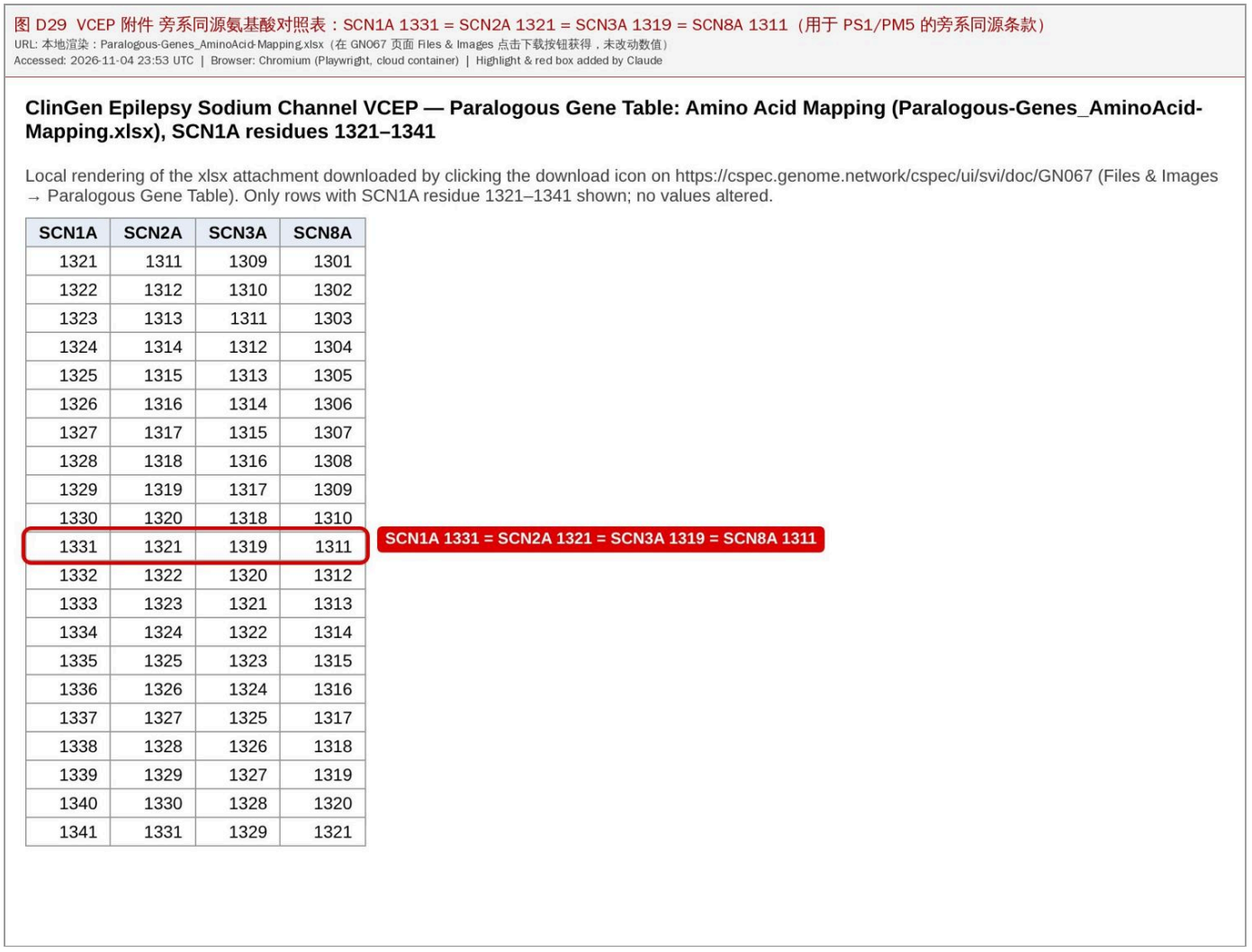
文件：screenshots/D27_VCEP_PS3功能实验阈值.jpg ｜ 来源：https://cspec.genome.network/cspec/ui/svi/doc/GN067 ｜ 访问：2026-11-04 23:53 UTC



PER #	SCN1A	SCN2A	SCN3A	SCN8A
10	1321	1311	1309	1301
10	1322	1312	1310	1302
10	1323	1313	1311	1303
10	1324	1314	1312	1304
10	1325	1315	1313	1305
10	1326	1316	1314	1306
10	1327	1317	1315	1307
10	1328	1318	1316	1308
10	1329	1319	1317	1309
10	1330	1320	1318	1310
10	1331	1321	1319	1311
10	1332	1322	1320	1312
10	1333	1323	1321	1313
10	1334	1324	1322	1314
10	1335	1325	1323	1315
10	1336	1326	1324	1316
10	1337	1327	1325	1317
10	1338	1328	1326	1318
10	1339	1329	1327	1319
10	1340	1330	1328	1320
10	1341	1331	1329	1321
10	1342	1332	1330	1322
10	1343	1333	1331	1323
10	1344	1334	1332	1324
10	1345	1335	1333	1325
10	1346	1336	1334	1326
10	1347	1337	1335	1327
10	1348	1338	1336	1328
10	1349	1339	1337	1329
10	1350	1340	1338	1330
10	1351	1341	1339	1331
10	1352	1342	1340	1332
10	1353	1343	1341	1333
10	1354	1344	1342	1334
10	1355	1345	1343	1335
10	1356	1346	1344	1336
10	1357	1347	1345	1337
10	1358	1348	1346	1338

图 D28　VCEP 附件 PM1 Table：SCN1A 1321–1364 属 PER #10；第 1331 行（SCN2A 1321 / SCN3A 1319 / SCN8A 1311）在表内 → PM1_Moderate 适用

文件：screenshots/D28_VCEP附件_PM1表_PER10.jpg ｜ 来源：本地渲染：PM1_Table_v2.20241205.xlsx（在 GN067 页面 Files & Images 点击下载按钮获得，未改动数值） ｜ 访问：2026-11-04 23:53 UTC



SCN1A	SCN2A	SCN3A	SCN8A
1321	1311	1309	1301
1322	1312	1310	1302
1323	1313	1311	1303
1324	1314	1312	1304
1325	1315	1313	1305
1326	1316	1314	1306
1327	1317	1315	1307
1328	1318	1316	1308
1329	1319	1317	1309
1330	1320	1318	1310
1331	1321	1319	1311
1332	1322	1320	1312
1333	1323	1321	1313
1334	1324	1322	1314
1335	1325	1323	1315
1336	1326	1324	1316
1337	1327	1325	1317
1338	1328	1326	1318
1339	1329	1327	1319
1340	1330	1328	1320
1341	1331	1329	1321

图 D29　VCEP 附件 旁系同源氨基酸对照表：SCN1A 1331 = SCN2A 1321 = SCN3A 1319 = SCN8A 1311（用于 PS1/PM5 的旁系同源条款）

文件：screenshots/D29_VCEP附件_旁系同源对照表.jpg　|　来源：本地渲染：Paralogous-Genes_AminoAcid-Mapping.xlsx（在 GN067 页面 Files & Images 点击下载按钮获得，未改动数值）　|　访问：2026-11-04 23:53 UTC

附录：只用接口或脚本核对、没有截图的项目

- NCBI E-utilities 下载 5 个 SCN1A RefSeq GenBank 记录并翻译（A5 补充说明）。
- VariantValidator REST：c.3993A>C 同样给出 p.(Glu1331Asp)，GRCh38 g.166009728T>G（网页版每个匿名用户提交次数有限，只截了 C>T 和 A>T）。
- gnomAD GraphQL API：区域内全部变异（含 AC0、AS_VQSR 等过滤变异）与逐位点覆盖度。
- ClinVar E-utilities：SCN1A 全部 5,556 条、SCN2A 3,203 条、SCN3A 2,272 条、SCN8A 2,664 条记录的标题筛选。
- PMC 开放获取全文接口：Pejaver 2022 表 2 的 REVEL 阈值。

待你在自己的 Chrome 补截的项目【需查】

1. NCBI RefSeq NM_001165963.4 页面（MANE Select 关键词、CDS 479..6508、ORIGIN 第 4441 行 “… ctcgatttga agggatgagg”）。
2. Pejaver 2022 表 2（PMC9748256）REVEL 行。
3. 如果报告里要放 ClinVar 网页截图（而不是 E-utilities 输出），在 ClinVar 打开 VCV002036225（SCN2A E1321Q）。

4. UCSC 保守性（可选；已有手动比对代替）。